

CountBCF: Bayesian Causal Forests for Count and Zero-Inflated Count Outcomes

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Abstract

Count outcomes – hospital admissions, infection cases, insurance claims, citation counts, recidivism events, single-cell read counts – are pervasive in the applied questions for which heterogeneous treatment effect (HTE) methods were designed, and a non-trivial fraction of these outcomes are also *zero-inflated*: some units never enter the count-generating process at all (a structural zero), while the rest accumulate strictly positive counts at heterogeneous intensities. A treatment can move the structural-zero probability and the count intensity independently, and the response-scale conditional average treatment effect (CATE) mixes the two channels. Existing tools collapse this structure: the workhorse Bayesian Causal Forest (BCF) of [Hahn et al. \[2020b\]](#) assumes a Gaussian response; Poisson and negative binomial generalised linear models, doubly robust extensions, and generalised random forests either rule out per-unit CATEs by design or discard BCF’s two structural ingredients (the differentially shrunk (μ, τ) decomposition and the propensity-as-covariate prognostic forest that combats regularization-induced confounding); classical zero-inflated regressions assume rigid treatment interactions; two-part learners separate the channels but discard the shared-covariate information. This paper introduces **CountBCF**, which embeds the BCF decomposition inside the log-linear BART backbone of [Murray \[2021\]](#). The framework covers two regimes within a single model: a *count* regime that delivers a (μ_f, τ_f) pair on the log-rate scale, and a *zero-inflated* regime that extends the decomposition to all three log-linear-BART forests – f, f_0, f_1 – yielding six forests in total. Each function group inherits the BCF differential-shrinkage philosophy and the propensity-as-covariate fix; posterior draws translate to per-unit response-scale CATEs in closed form, with a per-unit channel decomposition into count-rate and structural-zero effects available in the zero-inflated case. We release the `countbcf` R package, built on the `countbart` MCMC backend of [Wikle and Zigler \[2023\]](#), and pre-register a simulation study that crosses linear and nonlinear data-generating processes with Poisson, negative binomial, ZIP, and ZINB likelihoods and sweeps one-at-a-time over sample size, covariate dimension, and target average treatment effect.

Keywords: Bayesian causal forests; Conditional average treatment effects; Count regression; Heterogeneous treatment effects; Log-linear BART; Poisson and negative binomial regression; Zero-inflated regression; Structural zeros.

1 Introduction

Many of the outcomes that motivate modern causal inference are counts. Hospital admissions and emergency-department visits, infection incidences, dental caries, traffic crashes, insurance claims, citation totals, recidivism events, household child counts, and single-cell unique-molecular-identifier

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reads are all integer-valued and frequently bunched at zero. The applied literatures that produced these outcomes – epidemiology, health economics, ecology, operations research, finance, social statistics, computational biology – share an interest not just in whether an intervention works on average, but in *for whom* it works, and by how much: the conditional average treatment effect, or CATE. The past decade of heterogeneous treatment effect (HTE) methodology has answered that demand with flexible nonparametric tools – Bayesian Additive Regression Trees [Chipman et al., 2010, Hill, 2011], Bayesian Causal Forests [BCF; Hahn et al., 2020b], generalised random forests [Athey et al., 2019], meta-learners [Künzel et al., 2019], and double machine learning [Chernozhukov et al., 2018] – but the bulk of that methodology assumes a continuous, approximately Gaussian response.

Treating counts as if they were Gaussian has three well-documented failure modes [Cameron and Trivedi, 2013, Murray, 2021]. Count distributions place positive mass at zero and at small integers, often on a large fraction of the support; the conditional variance grows with the conditional mean, often super-linearly under overdispersion; and the substantive scientific quantity is almost always a rate or a relative risk, both of which live on the log scale. These pathologies bias point estimates, distort uncertainty quantification, and degrade the heterogeneity signal that HTE methods are designed to recover. A second, sharper failure occurs when the count outcome is also *zero-inflated*: a non-trivial fraction of units never enter the count-generating process at all (a structural zero), while the rest accumulate strictly positive counts at heterogeneous intensities. A canonical example is annual healthcare utilisation: in many provider panels and insurance cohorts, a sizeable fraction of enrollees record zero visits in a given year because they are effectively out of the care-seeking system, while the remaining enrollees record strictly positive visit counts at heterogeneous intensities. The same pattern arises in manufacturing defect counts [Lambert, 1992], automobile insurance claims, ecological species-abundance surveys, sentencing and recidivism counts, and single-cell RNA sequencing, where a large excess-zero mass reflects technical dropout on top of a count process.

What makes zero-inflation especially awkward for causal inference is that an intervention can move *both* mechanisms, and the two movements need not point in the same direction. Continuing the healthcare example: a workplace wellness programme aimed at preventive care could plausibly pull some enrollees *into* the system – shrinking the structural-zero fraction – while simultaneously *lowering* the visit intensity of those already engaged because preventive care substitutes for acute episodes. The response-scale CATE is then a mixture of the two channel effects; whether it ends up positive, negative, or near zero depends on which channel dominates for which unit. A method that ignores the structural-zero mechanism necessarily mis-attributes its movement: if the treatment shifts the structural-zero probability but leaves the count rate untouched, a Poisson-only working model has to absorb the entire change into the rate estimate, biasing it; the same mis-attribution flows in the opposite direction when the treatment moves the count rate but not the structural-zero probability.

Faced with a count outcome and a CATE question, an applied analyst today reaches for one of three default tools, each of which gives up something the analysis arguably needs. The first is a Gaussian HTE method, most commonly BCF or causal BART, applied to the raw counts as if they were continuous; this preserves the regularisation and uncertainty quantification practitioners trust, but commits all three Gaussian pathologies above. The second is a parametric Poisson, negative binomial, ZIP, or ZINB generalised linear model with treatment-by-covariate interactions [Cameron and Trivedi, 2013, Lambert, 1992, Greene, 1994], possibly stabilised by propensity weighting [Rosenbaum and Rubin, 1983] or a doubly robust correction [Chernozhukov et al., 2018]; this respects the count likelihood but recovers heterogeneity only through pre-specified interaction terms, defeating the nonparametric premise that motivated the move to HTE methods in the first place. The third is a generalised random forest with a Poisson moment condition [Athey et al., 2019], or a related relative-risk-targeted estimator [Kennedy, 2024]; this is honest about the count

structure but discards the two structural priors that distinguish BCF from other forest-based HTE methods, leaving heterogeneity estimates exposed to the high-variance behaviour that motivated those priors to begin with. Two-part and hurdle learners [Mullahy, 1986] separate the channels of a zero-inflated outcome but do not share covariate information across them and do not implement BCF differential shrinkage.

The structural priors in question are the distinguishing features of Hahn et al. [2020b]. Their construction decomposes the conditional mean of the response into a *prognostic* function – what the outcome would be in the absence of treatment, as a smooth function of the covariates – and a *moderating* function – by how much treatment shifts that baseline at each point in covariate space. The two functions are fit by separate BART ensembles with deliberately different priors: the prognostic forest is large and flexible; the moderating forest is small and tightly shrunk toward zero, reflecting the prior belief that treatment effect heterogeneity is typically smoother and weaker than the prognostic surface itself. This differential shrinkage stabilises CATE estimates in finite samples without forcing them to be constant. The second structural ingredient is the inclusion of an estimated propensity score among the inputs to the prognostic forest only. Hahn et al. [2020b] show that omitting this control creates *regularization-induced confounding* (RIC): the moderating forest absorbs prognostic signal that the regularised prognostic forest underfits, biasing CATE estimates in the direction of the propensity score. Both pathologies are at least as severe under a log-link count likelihood as under a Gaussian one: the multiplicative geometry of the log scale amplifies the consequences of prognostic underfitting, and the count likelihood’s tight mean-variance coupling leaves the moderating forest doubly exposed to RIC bias. The same arguments apply *per channel* in the zero-inflated case.

Our contribution. This paper develops **CountBCF**, a unified Bayesian causal forest framework for count and zero-inflated count outcomes that retains both of BCF’s structural ingredients while replacing its Gaussian backbone with a log-linear count likelihood. The framework covers two regimes within a single model and a single sampler:

1. **A count-faithful BCF.** For non-zero-inflated count outcomes, we extend the Bayesian Causal Forest of Hahn et al. [2020b] from a Gaussian to a Poisson and a negative binomial likelihood by routing both the prognostic and the moderating forest through the log-linear BART backbone of Murray [2021]. The result is a BCF that respects discreteness, mass at zero, and mean-variance coupling without abandoning the prognostic/moderating decomposition that defines the BCF family.
2. **A six-forest BCF decomposition for zero-inflated outcomes.** Murray’s log-linear BART parameterises a zero-inflated count likelihood through three function groups: the count log-rate f , the structural-zero log-odds f_0 , and the not-structural-zero log-odds f_1 , with the structural-zero probability identified as $\sigma(f_0 - f_1)$. We replace each group’s single forest with the BCF propensity-adjusted prognostic / moderator pair, yielding six forests in total: (μ_f, τ_f) for the count channel and $(\mu_{f_0}, \tau_{f_0}), (\mu_{f_1}, \tau_{f_1})$ for the ZI channel. The non-zero-inflated case is recovered as the three-forest sub-model.
3. **RIC protection on the log scale, per channel.** We retain the propensity-as-covariate fix and the differential-shrinkage prior structure of BCF, with the prognostic forest of each function group receiving more trees and a weaker zero-bias prior than the corresponding moderating forest. The leaf prior is the GIG mixture of Murray [2021], conjugate under a latent-variable augmentation; concentration hyperparameters are calibrated separately for the count and ZI groups, so each moderating forest is shrunk strictly more aggressively than its prognostic counterpart.

4. **Closed-form response-scale heterogeneity and channel decomposition.** The log-linear parameterisation means that, given posterior draws of the forests at any covariate value, the response-scale conditional treatment effect is available without any additional Monte Carlo step: it is a deterministic function of the stored forest values. We report per-unit posterior means, 50% and 95% credible intervals, and population-averaged ATE summaries, all on the response scale. In the zero-inflated case the same posterior draws produce a per-unit decomposition of the CATE into a count-rate channel $\Delta\lambda(x)$ and a structural-zero channel $\Delta p_{zi}(x)$.
5. **Open software and a pre-registered simulation study.** We release `countbcf`, an open-source R package built on the `countbart` MCMC backend of [Wikle and Zigler \[2023\]](#), with a single user-facing entry point that consumes a treated/control split and returns the posterior draw matrices needed for the closed-form CATE. The package is accompanied by a pre-registered simulation study (Section 6) that crosses linear and nonlinear data-generating processes with Poisson, NB, ZIP, and ZINB likelihoods and sweeps one-at-a-time over sample size, covariate dimension, and target average treatment effect, scoring all methods against a rate-scale truth.

The remainder of the paper proceeds as follows. Section 2 places CountBCF in the literature on count and zero-inflated causal inference and identifies the specific gap it fills. Section 3 fixes notation, including the latent structural-zero indicator and the response-scale potential-outcome means that are the paper’s target estimand. Section 4 reviews the log-linear BART machinery, the GIG-mixture leaf prior, and the redundant (f_0, f_1) parameterisation that makes the ZI leaf updates conditionally conjugate. Section 5 introduces CountBCF proper: the likelihood and parameterisation in both the count and the zero-inflated regimes, the differential-shrinkage prior structure, the augmented Gibbs sampler, and the closed-form response-scale CATE and channel-decomposition summaries. Section 6 pre-registers the simulation study used to evaluate CountBCF against a Gaussian-BCF baseline. Sections 7 and 8 are reserved for empirical results and discussion, to be written once the simulation study completes.

2 Relationship to previous literature

This section reviews the literatures that bear on conditional average treatment effect (CATE) estimation for count and zero-inflated count outcomes. The cell of the methodological grid in which CountBCF sits – causal $\times$ count-aware (with or without zero-inflation) $\times$ nonparametric Bayesian – is sparsely populated; the subsections that follow walk the adjacent cells in turn before Section 2.4 states the gap that CountBCF fills.

2.1 Causal inference for count outcomes: parametric and semiparametric eras

Causal analysis of count outcomes has its longest tradition in the parametric era of generalised linear models. The Poisson and negative binomial GLMs with a log link [[Cameron and Trivedi, 2013](#)] deliver interpretable rate-ratio summaries of the average causal effect under an unconfoundedness assumption and a correctly specified mean function. These models extend cleanly to the negative binomial when equi-dispersion fails, and when unconfoundedness is plausible only conditionally on covariates, the prescription is essentially the same as for continuous outcomes: condition on the covariates or on the propensity score [[Rosenbaum and Rubin, 1983](#)]. The two-stage residual inclusion of [Terza et al. \[2008\]](#) and the marginal structural models of [Robins et al. \[2000\]](#) relax

the unconfoundedness assumption in different ways while keeping the parametric log-linear-mean structure. Their common feature is that they target the *average* effect under a *fixed functional form*: heterogeneity, if permitted at all, enters through pre-specified treatment-by-covariate interactions, which is workable for a small number of binary moderators but not for a moderate-dimensional covariate vector with unknown interaction structure.

The semiparametric and doubly robust literature relaxes the parametric programme’s commitment to a single correct outcome model. Funk et al. [2011] survey doubly robust estimators across continuous and discrete outcomes, including Poisson outcome models paired with logistic propensity models. van der Laan and Gruber [2010] and van der Laan and Rose [2011] develop targeted maximum likelihood estimation (TMLE) as a general semiparametric framework whose construction extends naturally to count outcomes via a Poisson or negative binomial submodel for the targeted update step. Chernozhukov et al. [2018] introduce the double / debiased machine learning (DML) framework: the outcome regression and propensity score are fit by arbitrary high-quality machine learners, cross-fitted across folds to avoid overfitting bias, and combined in a Neyman-orthogonal score whose root yields an asymptotically normal ATE estimate. DML is outcome-model-agnostic: a Poisson-loss gradient boosting machine, a neural network, or BART can sit in the outcome slot without changing the orthogonality property. By design these methods target the average treatment effect rather than the per-unit CATE; heterogeneity, when reported at all, is typically delivered by stratifying across pre-specified subgroups or by a separate post-hoc fit of the outcome regression – neither of which preserves the orthogonality property that motivated the doubly robust construction. We adopt the spirit of the doubly robust programme – propensity adjustment of the outcome regression – via the route of Hansen [2008] and Hahn et al. [2020b], but reject the restriction to the average effect.

2.2 Classical and causal zero-inflated count models

The starting point for any zero-inflated analysis is the zero-inflated Poisson (ZIP) regression of Lambert [1992], which writes the response as a two-component mixture: with probability $\pi(x)$ the unit is a structural zero, and otherwise it is drawn from a Poisson with rate $\lambda(x)$. The hurdle alternative [Mullahy, 1986] factors the joint density into a binary “any vs. none” model and a zero-truncated count model; hurdle treats *all* zeros as structural while ZIP allows sampling zeros from the count component as well. Greene [1994] extends both constructions to the negative binomial case. Hall [2000] develops a random-effects ZIP for repeated-measures data, and Li [2012] settles the identifiability question: in the absence of constraints, the (π, λ) pair is not jointly identified from the marginal distribution of Y , but a logistic link on π together with a log link on λ and at least one continuous covariate restore identifiability. The `pscl` package of Zeileis et al. [2008] consolidates the count GLM toolkit – Poisson, NB, ZIP / ZINB, hurdle – in a single R treatment.

When interest shifts from describing Y to estimating a treatment effect, the two-component structure of ZI data complicates matters in two ways. First, the response-scale estimand is itself a sum of two interpretable contrasts: a shift in the structural-zero probability and a shift in the count rate. Second, identifying each contrast separately requires that the ignorability assumption be combined with the link-function restrictions of Li [2012] or with an instrument acting on only one channel. The naive plug-in strategy – estimate a ZIP / ZINB regression with treatment-covariate interactions and report the implied mean contrast – inherits all of the distributional rigidity of the underlying GLM. The two-part learner – fit a binary CATE for $\mathbf{1}\{Y_i > 0\}$ and a count CATE for $Y_i \mid Y_i > 0$, then combine – correctly separates the channels but loses joint efficiency across them and does not share covariate information. van der Laan and Rose [2011] discuss how doubly robust / TMLE constructions adapt to these working models; Kennedy [2024]

formalises the closely related mismatch between absolute-risk partitioning criteria and relative-risk targets. Guo et al. [2016] develop an IV mediation analysis for ZI counts that defines indirect effect *ratios* suitable for count outcomes. Yu et al. [2022] propose a Multiplicative Structural Nested Mean Model for ZI nonnegative longitudinal outcomes. Oganisian et al. [2021] build a Bayesian nonparametric model using Dirichlet process mixtures that jointly predict structural zeros, individualised treatment effects, and latent subgroups. Choi and Ni [2023] approach ZI outcomes from the causal-discovery direction. Lukusa et al. [2017] review the closely related literature on zero-inflated models with missing data. Each of these methods commits either to a particular longitudinal structural form, to a single response-scale estimand, or to a likelihood that does not isolate the structural-zero and count components for inference.

2.3 Heterogeneous-effects methods and the BART line

Methodological work on heterogeneous treatment effects has been dominated by Gaussian or binary outcomes. Hill [2011] introduced BART [Chipman et al., 2010, 1998] into the causal inference literature as a flexible learner for the outcome regression under unconfoundedness; her construction estimates a single BART model on (X, Z) and reads the CATE off as the difference of the model’s predictions at $Z = 1$ and $Z = 0$. This estimator inherits BART’s strong empirical performance and produces a fully Bayesian posterior over per-unit treatment effects, but it ties the prognostic and moderating components of the conditional mean to a single tree-structure prior, making no provision for the asymmetric regularisation that distinguishes heterogeneity estimation from outcome estimation, and – applied to a count outcome – it inherits the three Gaussian failure modes catalogued above.

Athey et al. [2019] introduce generalized random forests (GRF), a non-Bayesian alternative built on random forests with locally weighted moment conditions. GRF accommodates a Poisson moment for count outcomes; the asymptotic theory of Wager and Athey [2018] establishes pointwise consistency and asymptotic normality. GRF with a Poisson moment shares a feature CountBCF aims to escape: the splitting criterion targets local likelihood improvement, not relative-risk heterogeneity. Kennedy [2024] formalises this observation. Künzel et al. [2019] catalogue the standard meta-learners (S, T, X, DR), which are estimator-agnostic recipes that delegate the regression and propensity work to a user-chosen base. In principle a meta-learner can be deployed with a count-aware base, but the published meta-learner studies use Gaussian or binary base learners; meta-learners do not by themselves address RIC, and they offer no off-the-shelf solution for ZI counts.

The fourth strand – and the one from which CountBCF draws its computational backbone – is the line of work on BART and its count-aware extensions. Murray [2021] extends BART to count, binomial, and multinomial outcomes by replacing the Gaussian likelihood with a log-linear count likelihood (Poisson or negative binomial) and replacing the Gaussian leaf prior with a generalised inverse Gaussian (GIG) mixture leaf prior whose hyperparameters are calibrated so that, after exponentiation, the implied prior on the forest sum has zero mean and a user-controlled marginal variance. The mixture structure is engineered so that, after a Gamma latent-variable augmentation, the GIG mixture is conjugate. Section 4.2.2 of Murray [2021] extends the construction to zero-inflation through the *redundant* parameterisation $p_{zi,z}(x) = \sigma(f_0(x) - f_1(x))$, in which each of f_0 and f_1 carries its own log-linear BART prior, and the data drive symmetry-breaking through latent indicators. The redundancy is precisely what makes the leaf updates conditionally conjugate under a symmetric GIG-mixture prior. Wikle and Zigler [2023] package both constructions into the `countbart` R/C++ implementation that the present paper extends.

Hahn et al. [2020b] introduce Bayesian Causal Forests as the BART-based answer to the CATE-

estimation problem under unconfoundedness, with the differential-shrinkage and propensity-as-covariate ingredients catalogued in Section 1. The construction has antecedents in the prognostic score literature of Hansen [2008] and the RIC argument of Hahn et al. [2020a]. Wikle and Zigler [2023] restrict their count BART implementation to the predictive setting; they do not pursue the BCF decomposition. The combination of the BCF decomposition with the log-linear count likelihood (with or without zero-inflation), which is the content of the present paper, has not appeared in the literature.

2.4 Identified gap

To summarise: the predictive count and ZI-count cells are well populated, anchored by Murray [2021] and the `countbart` implementation of Wikle and Zigler [2023]. The causal Gaussian cell is well populated, anchored by Hahn et al. [2020b] on the Bayesian side and by generalized random forests [Athey et al., 2019, Wager and Athey, 2018] on the frequentist side. The causal count and causal ZI-count cells are sparsely populated, and within them no existing method combines (i) the propensity-adjusted (μ_f, τ_f) decomposition that has made BCF the de-facto Bayesian benchmark for heterogeneous causal effects, with (ii) a count-aware (and where applicable, zero-inflation-aware) likelihood that respects the discreteness, overdispersion, and structural-zero process of integer outcomes. The closest prior work either retains the Gaussian likelihood (BCF; Hahn et al., 2020b), drops the BCF decomposition (log-linear BART, with or without ZI; Murray, 2021), separates the two ZI channels and loses joint efficiency (two-part / hurdle learners; Mullahy, 1986; Guo et al., 2016; Yu et al., 2022; Oganisian et al., 2021), or replaces the Bayesian machinery with frequentist random forests targeting an absolute-risk or relative-risk criterion [Athey et al., 2019, Kennedy, 2024]. **CountBCF fills this gap.** It routes the prognostic and moderating forest of each log-linear BART function group through the backbone of Murray [2021] while preserving Hahn et al.’s propensity-adjusted differential-shrinkage prior structure, yielding a count-aware Bayesian causal forest with closed-form per-unit response-scale CATEs, a per-unit channel decomposition in the zero-inflated case, and full posterior uncertainty.

3 Problem statement and notation

Let $Y_i \in \mathbb{Z}_{\geq 0}$ denote a count-valued response, $Z_i \in \{0, 1\}$ a binary treatment indicator, and $X_i \in \mathbb{R}^p$ a length- p vector of pre-treatment covariates. We observe an iid sample $\{(Y_i, Z_i, X_i)\}_{i=1}^n$ from some joint distribution. Capital Roman letters denote random variables, with realisations in lower case. When Y or Z appear without a subscript they denote length- n column vectors; $\mathbf{X}$ is the $n \times p$ design matrix of covariates.

In the potential outcomes framework [Imbens and Rubin, 2015], write $Y_i(0), Y_i(1) \in \mathbb{Z}_{\geq 0}$ for the counterfactual responses under control and treatment, with observed outcome $Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$ (SUTVA). Throughout we assume *strong ignorability*:

$$Y_i(0), Y_i(1) \perp\!\!\!\perp Z_i \mid X_i, \tag{1}$$

$$0 < \pi(X_i) \equiv \Pr(Z_i = 1 \mid X_i) < 1, \tag{2}$$

the second of which (overlap) is required to identify treatment effects everywhere in covariate space. The estimated propensity score is denoted $\hat{\pi}(X_i)$; for the simulation study of Section 6 we set $\hat{\pi} = \pi$.

Latent structural-zero indicator. In the zero-inflated regime, the response splits into a structural-zero component and a count component. Let $S_i \in \{0, 1\}$ be the latent indicator that unit i is a

structural zero:

$$S_i = 1 \implies Y_i = 0, \quad S_i = 0 \implies Y_i \mid X_i, Z_i \sim \text{Poisson}(\lambda(X_i, Z_i)) \quad (\text{or } \text{NB}(\lambda, \kappa)), \quad (3)$$

and write the conditional structural-zero probability as

$$p_{\text{zi}, z}(X_i) \equiv \Pr(S_i = 1 \mid X_i, Z_i = z), \quad z \in \{0, 1\}. \quad (4)$$

The marginal count-component mean is $\lambda(x, z) = \mathbb{E}[Y_i \mid X_i = x, Z_i = z, S_i = 0]$; we parameterise $\log \lambda$ via a sum-of-trees prior in Section 5. The non-zero-inflated regime is recovered by setting $p_{\text{zi}, z} \equiv 0$ (no structural-zero process); equivalently, $S_i \equiv 0$ for all i .

Response-scale potential-outcome means. Combining the two channels yields the response-scale potential-outcome means

$$\mu_z(x) = \mathbb{E}[Y_i(z) \mid X_i = x] = (1 - p_{\text{zi}, z}(x)) \exp(\log \lambda_z(x)), \quad z \in \{0, 1\}. \quad (5)$$

Our target estimands are the *response-scale CATE* and ATE:

$$\tau(x) = \mu_1(x) - \mu_0(x), \quad \text{ATE} = \mathbb{E}_X[\tau(X)]. \quad (6)$$

When $p_{\text{zi}, z} \equiv 0$, (5) reduces to $\mu_z(x) = \exp(\log \lambda_z(x))$ and (6) becomes the plain count CATE.

Two-channel decomposition. In the zero-inflated regime, the response-scale CATE in (6) mixes a change in the count intensity and a change in the structural-zero probability. Methods that fit only one channel necessarily miss part of the effect. In Section 5.4 we therefore report – alongside τ – the two per-unit channel decompositions

$$\Delta \lambda(x) = \exp(\log \lambda_1(x)) - \exp(\log \lambda_0(x)), \quad \Delta p_{\text{zi}}(x) = p_{\text{zi}, 1}(x) - p_{\text{zi}, 0}(x). \quad (7)$$

Scales. Treatment effects naturally live on up to three scales:

- the *count log-rate scale*, on which the moderating forest $\tau_f(x)$ is parameterised; this is a relative-risk-like quantity, $\tau_f(x) \approx \log\{\lambda_1(x)/\lambda_0(x)\}$ in the non-zero-inflated case;
- the *ZI log-odds scale*, on which the treatment shift in η_{zi} is parameterised (zero-inflated regime only);
- the *response (rate) scale*, $\tau(x)$ in (6), against which all reported metrics are evaluated.

All relevant quantities are reported. The simulation study (Section 6) is scored against the rate-scale truth.

4 Background: log-linear BART

This section gives the condensed restatement of the log-linear BART construction of Murray [2021] – including the zero-inflation extension of §4.2.2 – that we need to define CountBCF in Section 5. The prognostic and moderating forests of each function group are individually log-linear BART regressions, so the leaf prior, the conditional sufficient statistics, and the latent-variable augmentation recapped here reappear unchanged inside the CountBCF Gibbs sampler. A reader already fluent in Murray [2021] §§3–4.2.2 may skip to Section 5.

4.1 The BART prior

A single BART regression [Chipman et al., 2010] expresses an unknown function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ as a sum of m shallow regression trees,

$$f(\mathbf{x}) = \sum_{h=1}^m g(\mathbf{x}; T_h, M_h), \quad (8)$$

where T_h is a binary tree partitioning $\mathbb{R}^p$ and $M_h = (\mu_{h,1}, \dots, \mu_{h,b_h})'$ collects the leaf parameters at the b_h terminal nodes of T_h ; the operator $g(\mathbf{x}; T_h, M_h)$ returns $\mu_{h,t}$ when $\mathbf{x}$ falls in terminal node t of T_h . The tree prior of Chipman et al. [1998] assigns each internal node at depth d a split probability $\alpha(1+d)^{-\beta}$, with $(\alpha, \beta) = (0.95, 2)$ biasing trees toward stubs. With a Gaussian response, a Gaussian leaf prior is conjugate, and the posterior is sampled by a backfitting Gibbs scheme that updates one tree at a time using grow / prune / change / swap MH moves on T_h and a conjugate draw of $M_h | T_h$.

4.2 The log-linear extension

For non-negative responses one writes the same sum-of-trees representation on the log scale,

$$\log f(\mathbf{x}) = \sum_{h=1}^m g(\mathbf{x}; T_h, M_h), \quad (9)$$

and reparameterises each leaf parameter as $\lambda_{h,t} = \exp(\mu_{h,t}) > 0$, so that $f(\mathbf{x}) = \prod_h \lambda_{h,t(\mathbf{x}; T_h)}$. The Poisson and negative binomial regression models take the form

$$Y_i | \mathbf{X}_i \sim \text{Poisson}(\mu_{0i} f(\mathbf{X}_i)), \quad Y_i | \mathbf{X}_i \sim \text{NB}(\text{size} = \kappa, \text{mean} = \mu_{0i} f(\mathbf{X}_i)), \quad (10)$$

with μ_{0i} a unit-specific offset and κ the NB dispersion, satisfying $\text{Var}(Y_i) = \mu_{0i} f(\mathbf{X}_i) + \mu_{0i}^2 f(\mathbf{X}_i)^2 / \kappa$. The Poisson model is the $\kappa \rightarrow \infty$ limit. Both likelihoods reappear inside the zero-inflated construction of Section 4.5: the not-structural-zero component of the ZI mixture is exactly the model in (10), while the structural-zero process is governed by an independent pair of log-linear BART components introduced below.

After collecting all but the h th tree's fit into $f^{-h}(\mathbf{X}_i) = \prod_{l \neq h} g(\mathbf{X}_i; T_l, \Lambda_l)$, the contribution of a single observation to the joint posterior over (T_h, Λ_h) has the conditionally Gamma-shaped form

$$L((T_h, \Lambda_h); T_{-h}, \Lambda_{-h}, y) \propto \prod_{t=1}^{b_h} \lambda_{h,t}^{r_{h,t}} \exp[-s_{h,t} \lambda_{h,t}], \quad (11)$$

with conditional sufficient statistics

$$r_{h,t} = \sum_{i: \mathbf{X}_i \in \mathcal{A}_{h,t}} u_i, \quad s_{h,t} = \sum_{i: \mathbf{X}_i \in \mathcal{A}_{h,t}} v_i f^{-h}(\mathbf{X}_i), \quad (12)$$

$\mathcal{A}_{h,t}$ the leaf- t partition cell of T_h , and (u_i, v_i) likelihood-specific summaries that depend on the function group (see Sections 4.4 and 4.5).

4.3 The GIG-mixture leaf prior

Murray [2021] constructs a leaf prior conjugate to (11) from two pieces. The generalised inverse Gaussian (GIG) family has density

$$p_{\text{GIG}}(\lambda \mid \eta, \chi, \psi) \propto \lambda^{\eta-1} \exp\left[-\frac{1}{2}(\chi/\lambda + \psi\lambda)\right], \quad \lambda > 0, \quad (13)$$

covering the Gamma ($\chi = 0$) and inverse-Gamma ($\psi = 0$) families as boundary cases. The CountBCF leaf prior is the equal mixture

$$p(\lambda_{h,t} \mid c, d) = \frac{1}{2} p_{\text{GIG}}(\lambda_{h,t} \mid -c, 2d, 0) + \frac{1}{2} p_{\text{GIG}}(\lambda_{h,t} \mid +c, 0, 2d), \quad (14)$$

with stochastic representation $\lambda_{h,t} = W$ with probability $1/2$ and $\lambda_{h,t} = 1/W$ otherwise, $W \sim \text{Gamma}(c, d)$. This construction makes $\log \lambda_{h,t}$ exactly symmetric about zero and keeps (14) conjugate to (11): the leaf full conditional is again a GIG mixture, and the integrated likelihood for the tree MH step is a ratio of GIG normalising constants [Murray, 2021, §3.1].

The pair (c, d) is reparameterised by a user-facing concentration $-a_0$ for the count component, z_{conc} for the ZI components (Section 4.5) – and the number of trees m , by setting $\psi_1(c) = a_0^2/m$ and $d = \exp(\psi(c))$, where ψ and ψ_1 are the digamma and trigamma functions. This calibration gives $\mathbb{E}[\log \lambda_{h,t}] = 0$ and $\text{Var}(\log \lambda_{h,t}) = a_0^2/m$, so the implied prior on $\log f(\mathbf{X}) = \sum_h \log \lambda_{h,t}(\mathbf{x}; T_h)$ is approximately $N(0, a_0^2)$ at any $\mathbf{X}$. The concentration therefore plays the same role for $\log f$ that the prior leaf scale plays for f in Gaussian BART. For large m or small a_0 the calibration admits the closed-form approximation $c \approx m/a_0^2 + 1/2$, $d \approx m/a_0^2$ [Murray, 2021, Proposition 3].

4.4 Latent-variable augmentation for the count component

To put the count likelihood into the form (11), Murray [2021] introduces one Gamma latent variable ξ_i per observation, with full conditional $\xi_i \mid - \sim \text{Gamma}(\kappa + y_i, \kappa + \mu_{0i}f(\mathbf{X}_i))$ under the NB likelihood, and $\xi_i \equiv 1$ under the Poisson likelihood. Conditioning on ξ_i , the per-observation contribution of f collapses to a Poisson-type kernel $f(\mathbf{X}_i)^{y_i} \exp[-\xi_i \mu_{0i}f(\mathbf{X}_i)]$, which integrated against the GIG-mixture prior (14) gives (11) with $u_i = y_i$ and $v_i = \xi_i \mu_{0i}$. Under the NB likelihood the dispersion κ is updated by a random-walk MH step on $\log \kappa$ with a Beta-prime prior implied by $\kappa/(1 + \kappa) \sim \text{Beta}(\kappa_a, \kappa_b)$. We defer the specific update to Section 5.3.

4.5 Zero-inflation and the redundant (f_0, f_1) parameterisation

For a zero-inflated count outcome the likelihood is a two-component mixture: with conditional probability $\omega(\mathbf{X}_i)$ the response is a structural zero, and otherwise it is drawn from the Poisson or NB law (10). Murray [2021] §4.2.2 writes the mixing probability through a pair of log-linear BART forests f_0, f_1 ,

$$\omega(\mathbf{X}_i) = \frac{\exp(f_0(\mathbf{X}_i))}{\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))} = \sigma(f_0(\mathbf{X}_i) - f_1(\mathbf{X}_i)), \quad (15)$$

where each of f_0, f_1 has its own log-linear BART prior (9) with an independent GIG-mixture leaf prior (14) (with concentration z_{conc}). The parameterisation is *redundant*: adding a common constant to both f_0 and f_1 leaves ω unchanged. The redundancy is identified through the prior – both forests are centred at zero with symmetric GIG-mixture leaf priors – and confers two concrete benefits. Conditionally Gamma-shaped leaf updates with closed-form GIG-mixture posteriors are obtained via a pair of latent variables. Let $z_i \in \{0, 1\}$ be the augmented mixture indicator that resolves whether an observed $Y_i = 0$ came from the structural process ($z_i = 0$) or from the count

process with a zero draw ($z_i = 1$). When $Y_i > 0$, $z_i \equiv 1$; when $Y_i = 0$, z_i has the Bernoulli full conditional

$$\Pr(z_i = 1 \mid -) = \frac{(1 - \omega(\mathbf{X}_i)) p_{\text{count}}(0 \mid \mathbf{X}_i)}{\omega(\mathbf{X}_i) + (1 - \omega(\mathbf{X}_i)) p_{\text{count}}(0 \mid \mathbf{X}_i)}, \quad (16)$$

where $p_{\text{count}}(0 \mid \mathbf{X}_i)$ is the count-component zero probability ($\exp(-\lambda)$ for Poisson / ZIP, $(\kappa/(\kappa + \lambda))^\kappa$ for NB / ZINB). Let further $\phi_i > 0$ be an Exponential latent decoupling f_0 and f_1 , with full conditional

$$\phi_i \mid - \sim \text{Exponential}(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))). \quad (17)$$

Conditioning on (z_i, ϕ_i) , the joint conditional likelihood of (f_0, f_1) factorises into independent terms of the form (11): the f_0 leaves have $(u_i, v_i) = (1 - z_i, \phi_i)$ and the f_1 leaves have $(u_i, v_i) = (z_i, \phi_i)$ [Murray, 2021, §A.5, Proposition 1]. The redundant sampler is also numerically better behaved than a strictly identified parameterisation: Murray [2021, Supplement §A] report substantially better effective-sample-size performance across the full probability range. In the body of this paper we write the structural-zero log-odds as $\eta_{zi} = f_0 - f_1$ and report inference for the identified functionals only.

4.6 Implementation backbone

We inherit the C++ implementation of the log-linear BART sampler – the GIG-mixture leaf draw, the grow / prune / change / swap tree MH moves, the latent-variable updates (z_i, ξ_i, ϕ_i) , the soft-mixture-weight recomputation, and the dispersion MH step – from the `countbart` R package of Wikle and Zigler [2023]. CountBCF adds the BCF (μ_f, τ_f) decomposition on top of this backend by replacing the single forest of log-linear BART (per function group) with a forest pair and introducing a per-observation weight that routes treated and control units to the correct forest; we describe that change next.

5 CountBCF: BCF decomposition for count and ZI count outcomes

This section defines CountBCF. Section 5.1 states the likelihood and parameterisation in both the count and the zero-inflated regimes; Section 5.2 collects the priors and the differential shrinkage applied symmetrically across function groups; Section 5.3 writes out the MCMC sweep in Algorithm 1 and the associated Gibbs and MH updates; Section 5.4 explains how posterior draws are post-processed into the estimands of Section 3, including the per-unit channel decomposition in the ZI case.

5.1 Likelihood and parameterisation

CountBCF models the conditional mean of a count response (and, in the ZI regime, the structural-zero probability) on the log scale using a treatment-aware BCF decomposition. We treat the count and zero-inflated cases as two sub-models of a common framework, selected by the user via a `count_model` switch that takes values in `{poisson, nb, zipoisson, zinb}`.

Count regime: three forests. For unit i with covariates X_i and treatment indicator $Z_i \in \{0, 1\}$, the Poisson / NB sub-model writes

$$\log \lambda(X_i, Z_i) = \mu_f(X_i, \hat{\pi}(X_i)) + Z_i \cdot \tau_f(X_i), \quad (18)$$

Table 1: The six forests of CountBCF in the zero-inflated regime. The function group g labels the log-linear BART target inherited from Murray [2021]; the BCF role s labels the prognostic (μ) vs. moderating (τ) piece introduced by the (μ, τ) decomposition. The non-ZI sub-model retains only the $g = 0$ row.

g	Function group	Forest pair $(\mu^{(g)}, \tau^{(g)})$	Trees (m_μ, m_τ)	Weight ω_i on $\tau^{(g)}$
0	Count log-rate log λ	(μ_f, τ_f)	(250, 50)	Z_i
1	ZI log-odds f_0	(μ_{f_0}, τ_{f_0})	(100, 50)	Z_i
2	not-ZI log-odds f_1	(μ_{f_1}, τ_{f_1})	(100, 50)	Z_i

with response

$$Y_i \mid X_i, Z_i \sim \text{Poisson}(\lambda(X_i, Z_i)) \text{ or } \text{NB}(\text{size} = \kappa, \text{mean} = \lambda(X_i, Z_i)), \quad (19)$$

where the size parameter $\kappa > 0$ governs overdispersion. The Poisson model is the $\kappa \rightarrow \infty$ limit. Decomposition (18) is the count analogue of the additive BCF decomposition $\mathbb{E}[Y \mid X, Z] = \mu_f(X, \hat{\pi}) + Z\tau_f(X)$ of Hahn et al. [2020b]. Both pieces are log-linear BART forests in the sense of Section 4.2: μ_f is a sum of m_μ trees with leaf parameters drawn from the GIG mixture (14), and similarly τ_f is a sum of m_τ trees. The prognostic forest models the control-arm log-mean across the whole sample; the moderating forest enters only for treated units and captures treatment-effect heterogeneity. As in BCF, $\hat{\pi}$ is appended to the covariates feeding μ_f but *not* those feeding τ_f .

Zero-inflated regime: six forests, three groups. The ZIP / ZINB sub-model attaches a BCF (μ, τ) decomposition to each of the three log-linear BART forests f, f_0, f_1 of Section 4.5. We index the three function groups by $g \in \{0, 1, 2\}$ with $g = 0$ the count log-rate, $g = 1$ the structural-zero log-odds component f_0 , and $g = 2$ the not-structural-zero log-odds component f_1 . Then

$$\begin{aligned} \log \lambda(X_i, Z_i) &= \mu_f(X_i, \hat{\pi}(X_i)) + Z_i \cdot \tau_f(X_i), \\ \eta_{\text{zi}}(X_i, Z_i) &= [\mu_{f_0}(X_i, \hat{\pi}(X_i)) + Z_i \cdot \tau_{f_0}(X_i)] \\ &\quad - [\mu_{f_1}(X_i, \hat{\pi}(X_i)) + Z_i \cdot \tau_{f_1}(X_i)], \\ p_{\text{zi}, z}(X_i) &= \sigma(\eta_{\text{zi}}(X_i, Z_i = z)), \quad z \in \{0, 1\}, \end{aligned} \quad (20)$$

together with the ZI count likelihood $Y_i \mid S_i = 1 = 0$, and $Y_i \mid S_i = 0, X_i, Z_i \sim \text{Poisson}(\lambda(X_i, Z_i))$ or $\text{NB}(\kappa, \lambda(X_i, Z_i))$, where $\Pr(S_i = 1 \mid X_i, Z_i) = p_{\text{zi}, z}(X_i)$. The non-ZI sub-model collapses (20) by dropping the η_{zi} line and the two ZI groups, recovering (18). Equation (20) contains six unknown functions, two per function group, summarised in Table 1. Each is itself a log-linear BART regression: a sum of $m_{s,g}$ shallow trees with leaf parameters drawn from the GIG mixture, with $s \in \{\mu, \tau\}$ indexing the BCF role and $g \in \{0, 1, 2\}$ indexing the function group. For each group, the prognostic forest $\mu^{(g)}$ enters with weight $\omega_i = 1$; the moderating forest $\tau^{(g)}$ enters with weight $\omega_i = Z_i$. The propensity score estimate $\hat{\pi}$ is appended to the covariates of all three prognostic forests but not to the moderating ones.

Scales recovered. On the *count log-rate scale*, $\tau_f(x)$ is itself a CATE: the conditional log relative rate (of the not-structural-zero sub-population in the ZI case), $\tau_f(x) = \log\{\lambda_1(x)/\lambda_0(x)\}$. On the *ZI log-odds scale*, the treatment shift in the structural-zero log-odds is

$$\Delta\eta_{\text{zi}}(x) \equiv \eta_{\text{zi}}(x, 1) - \eta_{\text{zi}}(x, 0) = \tau_{f_0}(x) - \tau_{f_1}(x), \quad (21)$$

so τ_{f_0}, τ_{f_1} are jointly identified up to a common shift – only the difference is statistically relevant for the ZI channel. On the *response (rate) scale*, the potential-outcome means (5) are

$$\mu_z(x) = (1 - \sigma(\eta_{zi}(x, z))) \exp(\mu_f(x, \hat{\pi}(x)) + z\tau_f(x)), \quad z \in \{0, 1\}, \quad (22)$$

and $\tau(x) = \mu_1(x) - \mu_0(x)$. In the non-ZI sub-model $p_{zi,z} \equiv 0$ and (22) reduces to $\mu_z(x) = \exp(\mu_f + z\tau_f)$, giving the closed-form count CATE

$$\tau(x) = \exp(\mu_f(x, \hat{\pi}(x))) (\exp(\tau_f(x)) - 1), \quad (23)$$

which makes explicit that the absolute treatment effect on the rate scale depends on the baseline rate in addition to the log-rate effect. In the ZI case, τ also mixes the count-rate channel $\Delta\lambda$ and the ZI-probability channel Δp_{zi} from (7).

Why this parameterisation. Three properties of the construction are worth flagging. First, the model respects the non-negativity of Y automatically: $\lambda > 0$ and (where applicable) $p_{zi,z} \in (0, 1)$ regardless of the forest values, so no truncation or clipping is required. Second, all forests are parameterised *a priori* independently, which is what permits the BCF differential-shrinkage idea [Hahn et al., 2020b, §2.3] to be applied per group: each $\tau^{(g)}$ can be shrunk harder than its $\mu^{(g)}$ counterpart without disturbing other groups’ priors. Third, the construction reduces to log-linear BART [Murray, 2021] when $\tau^{(g)} \equiv 0$ for all g , and to standard BCF in the limit where τ_f is shrunk hard to zero. In this sense CountBCF is the smallest extension of log-linear BART that delivers a separate moderating forest on each channel, and the smallest extension of BCF that handles count and zero-inflated count outcomes natively. The redundant (f_0, f_1) parameterisation in the ZI case is retained for the reasons given in Section 4.5: conjugate leaf updates, well-defined posterior on the identified functionals, and substantially better MCMC mixing [Murray, 2021, Supplement §A]. The BCF (μ, τ) decomposition does not break the redundancy: shifting μ_{f_0} and μ_{f_1} by a common constant leaves η_{zi} invariant, and shifting τ_{f_0} and τ_{f_1} by a common constant leaves $\Delta\eta_{zi}$ invariant.

5.2 Priors

The CountBCF prior factorises across forests, the dispersion parameter, and the GIG-mixture latent structure of each leaf. The construction is symmetric across function groups, so we state the prior for a generic forest pair $(\mu^{(g)}, \tau^{(g)})$ and specialise the concentration hyperparameters to the count and ZI groups.

Sum-of-trees priors on each forest. Each forest is a log-linear BART regression in the sense of Section 4.2: a sum of $m_{s,g}$ shallow trees with tree structures $\{T_{s,g,h}\}$ drawn *a priori* independently from the CGM tree-structure prior and leaf parameters $\{\Lambda_{s,g,h}\} = \{\lambda_{s,g,h,t}\}$ drawn *a priori* independently from the GIG mixture (14). Collecting these, for $s \in \{\mu, \tau\}$ and (in the ZI case) $g \in \{0, 1, 2\}$,

$T_{s,g,h} \mid \alpha_s, \beta_s \stackrel{\text{iid}}{\sim} \text{CGM}(\alpha_s, \beta_s), \quad h = 1, \dots, m_{s,g},$	$\lambda_{s,g,h,t} \mid c_{s,g}, d_{s,g} \stackrel{\text{iid}}{\sim} \frac{1}{2} \text{GIG}(-c_{s,g}, 2d_{s,g}, 0) + \frac{1}{2} \text{GIG}(c_{s,g}, 0, 2d_{s,g}), \quad t = 1, \dots, b_{s,g,h},$
$\mu^{(g)}(x, \hat{\pi}(x)) = \sum_{h=1}^{m_{\mu,g}} g(x, \hat{\pi}(x); T_{\mu,g,h}, \Lambda_{\mu,g,h}),$	$\tau^{(g)}(x) = \sum_{h=1}^{m_{\tau,g}} g(x; T_{\tau,g,h}, \Lambda_{\tau,g,h}),$

(24)

Table 2: CountBCF default hyperparameters. The ZI rows ($g \in \{1, 2\}$) apply only when `count_model` $\in \{\text{zipoisson}, \text{zinb}\}$; the dispersion parameters apply only when `count_model` $\in \{\text{nb}, \text{zinb}\}$.

Hyperparameter	Default	Role
$m_{\mu,0}, m_{\tau,0}$	250, 50	Trees per forest, count group $g = 0$
$m_{\mu,1}, m_{\tau,1}$	100, 50	Trees per forest, ZI group $g = 1$ (f_0)
$m_{\mu,2}, m_{\tau,2}$	100, 50	Trees per forest, ZI group $g = 2$ (f_1)
(α_μ, β_μ)	(0.95, 2)	CGM depth penalty, all $\mu^{(g)}$
$(\alpha_\tau, \beta_\tau)$	(0.25, 3)	CGM depth penalty, all $\tau^{(g)}$
a_0	$\frac{1}{2}(\log y_{(0.95)} - \overline{\log \mu_{0,i}})$	Leaf concentration, μ_f (count)
$a_{0,\tau}$	$a_0/2$	Leaf concentration, τ_f (count)
z_{conc}	$3.5/\sqrt{2}$	Leaf concentration, μ_{f_0}, μ_{f_1} (ZI)
$z_{\text{conc},\tau}$	$z_{\text{conc}}/2$	Leaf concentration, τ_{f_0}, τ_{f_1} (ZI)
(κ_a, κ_b)	(5, 3)	Beta-prime prior on κ
σ_κ	0.21	RW-MH proposal SD on $\log \kappa$
<code>include_pihat</code>	"control"	Propensity appended to all $\mu^{(g)}$

where the CGM density assigns an internal-node split probability $\alpha_s (1 + d)^{-\beta_s}$ at depth d and a uniform draw over splitting variable and threshold given the split. The depth-penalty hyperparameters (α_s, β_s) are shared across function groups (so all prognostic forests have the same tree-shape prior, and similarly for the moderating forests); the leaf concentration is allowed to differ across g to reflect the different natural scales of the count log-rate versus the zero-inflation log-odds. In the count-only regime, indices $g \in \{1, 2\}$ are dropped throughout.

Differential shrinkage applied symmetrically. CountBCF inherits the BCF [Hahn et al., 2020b] idea that the treatment-effect surface $\tau^{(g)}$ should be shrunk more aggressively than the prognostic surface $\mu^{(g)}$ on *each* channel: the signal-to-noise in $\tau^{(g)}$ is typically lower than in $\mu^{(g)}$; spurious heterogeneity in any of $\tau_f, \tau_{f_0}, \tau_{f_1}$ manifests as fake interactions with Z on the respective channel; and a stub-biased prior on every $\tau^{(g)}$ delivers a near-constant treatment-effect estimator as a default. The default hyperparameters are summarised in Table 2. The tree-prior $(\alpha_\tau, \beta_\tau) = (0.25, 3)$ sharply downweights non-stub trees in every moderating forest: under it the prior probability of a non-trivial split at the root is 0.25 and the probability of a depth-2 internal node is approximately 0.0039, so on average each moderating forest is close to a piecewise-constant surface with very few splits. The leaf concentration $a_{0,\tau} = a_0/2$ (count group) and $z_{\text{conc},\tau} = z_{\text{conc}}/2$ (ZI groups) further shrinks the amplitude of those few splits: under the GIG-mixture calibration of Section 4.3, $\text{Var}(\log \lambda_{\tau,g,h,t}) \approx a_{0,\tau,g}^2/m_{\tau,g}$, so the implied marginal prior on $\tau^{(g)}(x)$ is $N(0, a_{0,\tau,g}^2)$, a factor of two tighter than the prior on $\mu^{(g)}$.

The user-facing concentration a_0 defaults to $\frac{1}{2}(\log y_{(0.95)} - \overline{\log \mu_{0,i}})$, where $y_{(0.95)}$ is the empirical 95% quantile of y , centring the count-channel prior on a data-matched scale. The ZI concentration z_{conc} defaults to $3.5/\sqrt{2} \approx 2.47$, matching the calibration of Murray [2021, §4.2.2] for the redundant (f_0, f_1) parameterisation. Both can be overridden by the user.

Propensity-as-covariate fix on every channel. A central insight of Hahn et al. [2020b] is that, on the additive response scale, naively regressing the outcome on (X, Z) with a BART-like prior leaves the estimator vulnerable to regularisation-induced confounding (RIC): when the propensity

score $\pi(x)$ is a function of x , the prior shrinkage of the prognostic surface toward zero biases τ_f in a direction that depends on π . The fix is to append $\hat{\pi}(x)$ to the prognostic forest's covariates so that μ_f can “absorb” the part of the outcome explained by π without recruiting τ_f . On the log-rate scale of (18) the RIC argument is at least as severe: a perturbation δ in μ_f translates to a multiplicative perturbation $e^\delta - 1 \approx \delta$ in the baseline rate, which then multiplies into τ via (23); a bias in μ_f correlated with π propagates linearly into τ . In the ZI case the same RIC argument applies *independently* on each channel: perturbations in μ_{f_0} or μ_{f_1} propagate into the ZI-channel moderating forests through the redundant parameterisation. We therefore append $\hat{\pi}$ to the covariate set of *every* prognostic forest by default and do *not* append it to any moderating forest. The R wrapper exposes this as `include_pihat = "control"`, with `"moderate"`, `"both"`, and `"none"` available for sensitivity checks.

Negative-binomial dispersion κ . Under the NB / ZINB likelihood, the size parameter κ controls overdispersion: $\text{Var}(Y_i \mid X_i, Z_i, S_i = 0) = \lambda_i + \lambda_i^2 / \kappa$. Following Murray [2021] we place a Beta-prime prior on κ implied by

$$\frac{\kappa}{1 + \kappa} \sim \text{Beta}(\kappa_a, \kappa_b), \quad (\kappa_a, \kappa_b) = (5, 3), \quad (25)$$

with mean 2.5 and a right tail heavy enough not to rule out near-Poisson behaviour while still penalising arbitrarily large overdispersion. The same prior governs κ under both NB and ZINB; structural zeros enter the κ -conditional likelihood only through z_i (Section 5.3). The dispersion is updated by a random-walk MH step on $\log \kappa$ with Gaussian proposal of standard deviation $\sigma_\kappa = 0.21$.

5.3 Posterior computation

Posterior inference for CountBCF uses a Gibbs sampler that interleaves backfitting updates of the forests with latent-variable augmentation steps and (under NB / ZINB) a MH step on the dispersion. The structure mirrors the log-linear BART sampler of Murray [2021, §3.3, §S.3]: the only forest-level change is the introduction of a per-observation weight $\omega_i \in \{1, Z_i\}$ that routes treated and control units to the moderating and prognostic forests, replicated across all function groups.

Latent-variable augmentation. The augmentations are inherited from Sections 4.4 and 4.5. Each observation carries up to three latents:

$$\begin{aligned} z_i \mid - &\sim \text{Bernoulli}\left(\frac{(1 - \omega(\mathbf{X}_i)) p_{\text{count}}(0 \mid \mathbf{X}_i)}{\omega(\mathbf{X}_i) + (1 - \omega(\mathbf{X}_i)) p_{\text{count}}(0 \mid \mathbf{X}_i)}\right) \quad \text{when } Y_i = 0 \text{ (ZIP/ZINB); } z_i \equiv 1 \text{ when } Y_i > 0, \\ \xi_i \mid - &\sim \begin{cases} \delta_1 & (\text{Poisson / ZIP}), \\ \text{Gamma}(\kappa + Y_i, \kappa + \lambda(\mathbf{X}_i, Z_i)) & (\text{NB / ZINB}), \end{cases} \\ \phi_i \mid - &\sim \text{Exponential}(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))) \quad (\text{ZIP/ZINB only}). \end{aligned} \quad (26)$$

Under the non-ZI sub-models the z_i and ϕ_i updates are skipped; under the Poisson / ZIP sub-models the ξ_i update is skipped.

Tree updates: a single primitive applied to all forests. Conditional on the latents, the joint conditional likelihood of the forests factorises into independent terms, each of which has

Table 3: Per-group sufficient statistics. The values u_i, v_i are inserted into (27) through the leaf accumulators $r_{s,g,h,t}, s_{s,g,h,t}$. The BCF weight is $\omega_i = 1$ for prognostic forests ($s = \mu$) and $\omega_i = Z_i$ for moderating forests ($s = \tau$); it premultiplies (u_i, v_i) before the leaf sum is formed. The non-ZI sub-model retains only the $g = 0$ row.

g	Function target	$u_i \cdot \omega_i^{-1}$	$v_i \cdot \omega_i^{-1}$
0	count rate λ	$z_i y_i$	$z_i \exp(\log \xi_i + \log \lambda(\mathbf{X}_i, Z_i) - \omega_i \log \lambda_{s,g,h,t})$
1	ZI log-odds f_0	$1 - z_i$	$\exp(\log \phi_i + \log f_0(\mathbf{X}_i) - \omega_i \log \lambda_{s,g,h,t})$
2	ZI log-odds f_1	z_i	$\exp(\log \phi_i + \log f_1(\mathbf{X}_i) - \omega_i \log \lambda_{s,g,h,t})$

the conditionally Gamma form (11). For forest s in function group g at tree h , the conditional augmented likelihood for the $b_{s,g,h}$ leaves of $T_{s,g,h}$ is

$$L(\Lambda_{s,g,h} \mid T_{s,g,h}, T_{-(s,g,h)}, \Lambda_{-(s,g,h)}, z_i, \xi_i, \phi_i, Y, \kappa) \propto \prod_{t=1}^{b_{s,g,h}} \lambda_{s,g,h,t}^{r_{s,g,h,t}} \exp[-s_{s,g,h,t} \lambda_{s,g,h,t}], \quad (27)$$

where the leaf sufficient statistics $r_{s,g,h,t} = \sum_{i \in \mathcal{A}_{s,g,h,t}} u_i$ and $s_{s,g,h,t} = \sum_{i \in \mathcal{A}_{s,g,h,t}} v_i f^{-(s,g,h)}(\mathbf{X}_i)$ aggregate over the units in the partition cell, using the per-group (u_i, v_i) enumerated in Table 3. The BCF weight $\omega_i \in \{1, Z_i\}$ enters multiplicatively in both u_i and v_i : for prognostic forests every unit contributes; for moderating forests only treated units contribute. This is the BCF moderating-forest ingredient transposed to the count (or ZI count) likelihood, applied symmetrically to all function groups present in the model.

The GIG-mixture leaf prior (14) is conjugate to (27), so the leaf draw is a closed-form mixture of GIG variates (implemented in `drmu_loglinear`), and $T_{s,g,h}$ is updated by a MH step with grow / prune / change / swap proposals (implemented in `bd_loglinear`). The MH acceptance ratio uses the integrated marginal likelihood obtained by integrating $\Lambda_{s,g,h}$ against the GIG-mixture prior; because both prior and likelihood are products over leaves, that integral factorises and reduces to a ratio of GIG normalising constants [Murray, 2021, Appendix A].

Per-group log-fit accumulator. Implemented as backfitting, the tree update maintains running accumulators on the n -vector index k : `allfit`[k] tracks the total log-mean / log-odds; `allfits`[s][k] tracks the contribution of forest s ; and the per-group accumulator `log_f_per_group`[g][k] = $\mu^{(g)}(\mathbf{X}_k, \hat{\pi}(\mathbf{X}_k)) + Z_k \tau^{(g)}(\mathbf{X}_k)$ stores the in-sample log-fit of both forests in group g . Before sampling tree (s, g, h) its current contribution is subtracted from all relevant accumulators; sufficient statistics are computed from the partial fit; $T_{s,g,h}$ and $\Lambda_{s,g,h}$ are updated; and the new contribution is added back. The per-group accumulator makes the six-forest update efficient: the ZI sufficient statistics read $\log f_0$ and $\log f_1$ from `log_f_per_group`[1] and `log_f_per_group`[2] in $O(1)$ per observation. After each sweep the soft mixture weight that enters the z_i update is recomputed as $\log w(\mathbf{X}_i) = \log f_1(\mathbf{X}_i) - \log \text{sumexp}(\log f_0(\mathbf{X}_i), \log f_1(\mathbf{X}_i))$, which avoids the underflow of naive computation when one of the log-fits is large and negative.

Dispersion update (NB / ZINB only). Propose $\kappa^* = \exp(\log \kappa + \sigma_\kappa \varepsilon)$ with $\varepsilon \sim N(0, 1)$. Accept with probability

$$\alpha_\kappa = \min \left\{ 1, \frac{p_{\text{NB}}(Y \mid z_i, \lambda, \kappa^*)}{p_{\text{NB}}(Y \mid z_i, \lambda, \kappa)} \cdot \frac{p(\kappa^*)}{p(\kappa)} \cdot \frac{\kappa^*}{\kappa} \right\}, \quad (28)$$

where the likelihood factor is the augmented NB likelihood conditional on the mixture indicator (so structural zeros do not enter the dispersion update under ZINB), $p(\kappa)$ is the Beta-prime density implied by (25), and the Jacobian factor κ^*/κ comes from the log-scale proposal. Acceptance rates after burn-in are returned as `kappa_acpt` and serve as a coarse convergence diagnostic; we target the 0.20–0.45 band typical for random-walk MH.

Implementation lineage. The C++ primitives `bd_loglinear`, `drmu_loglinear`, `fit_loglinear`, `drz_loglinear`, `drxi_loglinear`, `drphi_loglinear`, and the dispersion MH step are inherited verbatim from `countbart` [Wikle and Zigler, 2023]; the CountBCF contribution is the dispatch logic in `src/countbcf.cpp` that (i) instantiates two forests per function group rather than one, (ii) sets the per-forest weight vector ω_i to either $\mathbf{1}_n$ or Z_{trt} , (iii) applies ω_i to the sufficient-statistic construction in step 5.2, and (iv) maintains the per-group log-fit accumulator so that the four ZI forests read f_0 and f_1 in $O(1)$ per observation. The change is mechanical at the kernel level but yields a sampler that, to our knowledge, is functionally distinct from any log-linear BART variant in the published literature: it jointly estimates up to six log-rate / log-odds surfaces under differential shrinkage, with each moderating forest seeing only the treated subsample’s residual variation.

5.4 Posterior summaries

Algorithm 1 produces a sequence of B post-burn-in posterior draws indexed by $b = 1, \dots, B$. The sampler stores, for each draw, the per-unit values of the active forests on the original (unsorted) input order:

$$\begin{aligned}\mu_f^{(b)}(X_i) &= \text{mu_f_post}[b, i], & \tau_f^{(b)}(X_i) &= \text{tau_f_post}[b, i], \\ \mu_{f_0}^{(b)}(X_i) &= \text{mu_f0_post}[b, i], & \tau_{f_0}^{(b)}(X_i) &= \text{tau_f0_post}[b, i], \\ \mu_{f_1}^{(b)}(X_i) &= \text{mu_f1_post}[b, i], & \tau_{f_1}^{(b)}(X_i) &= \text{tau_f1_post}[b, i],\end{aligned}\tag{29}$$

together with the dispersion draws $\kappa^{(b)}$ (NB / ZINB only). In the non-ZI sub-model only the first row is stored. Downstream inference is a post-processing exercise on these matrices and produces three (count) or four (ZI) classes of summaries.

Per-iteration potential outcomes. For each draw b and each unit i , the closed-form mapping from stored forest values to response-scale potential-outcome means is, for $z \in \{0, 1\}$,

$$\begin{aligned}\log \lambda_z^{(b)}(X_i) &= \mu_f^{(b)}(X_i) + z \cdot \tau_f^{(b)}(X_i), \\ \eta_{\text{zi}}[z]^{(b)}(X_i) &= [\mu_{f_0}^{(b)}(X_i) + z \cdot \tau_{f_0}^{(b)}(X_i)] - [\mu_{f_1}^{(b)}(X_i) + z \cdot \tau_{f_1}^{(b)}(X_i)], \\ \mu_z^{(b)}(X_i) &= (1 - \sigma(\eta_{\text{zi}}[z]^{(b)}(X_i))) \exp(\log \lambda_z^{(b)}(X_i)).\end{aligned}\tag{30}$$

In the non-ZI sub-model $\sigma(\eta_{\text{zi}}) \equiv 0$ and (30) collapses to $\mu_z^{(b)}(X_i) = \exp(\mu_f^{(b)}(X_i) + z \tau_f^{(b)}(X_i))$. The pair $(\mu_0^{(b)}, \mu_1^{(b)})$ collects the response-scale potential-outcome means at every unit under every draw, and all reported quantities below are deterministic functions of these two matrices – no additional sampling is required: posterior uncertainty in the response-scale CATE arises from joint uncertainty in the forests and is automatically propagated by the backfitting Gibbs sampler.

Response-scale CATE. The primary estimand is (6). The b th posterior draw is

$$\widehat{\tau}_{,i}^{(b)} = \mu_1^{(b)}(X_i) - \mu_0^{(b)}(X_i), \quad (31)$$

with the count-only form $\widehat{\tau}_{,i}^{(b)} = \exp(\mu_f^{(b)}(X_i))(\exp(\tau_f^{(b)}(X_i)) - 1)$ recovered in the non-ZI sub-model. The per-unit posterior mean and 95% equal-tailed credible interval are $\overline{\tau}_{,i} = B^{-1} \sum_b \widehat{\tau}_{,i}^{(b)}$ and $[Q_{0.025}, Q_{0.975}]$ across draws, with Q_q the empirical q -quantile. The log-rate CATE $\widehat{\tau}_{f,i}^{(b)} = \tau_f^{(b)}(X_i)$ is reported alongside for users whose applied questions are best phrased as relative rates.

Sample-average treatment effect. The ATE (6) is the in-sample mean of the response-scale CATE,

$$\widehat{\text{ATE}}^{(b)} = \frac{1}{n} \sum_{i=1}^n \widehat{\tau}_{,i}^{(b)}, \quad (32)$$

giving a B -vector of posterior draws. The posterior mean and 95% equal-tailed interval of this vector give the point estimate and uncertainty of the sample ATE. The average is over the empirical distribution of X , so (32) estimates the *sample-average* treatment effect rather than a population-average one; Hahn et al. [2020b] discuss why this is the natural target for BART-family methods.

Channel decomposition (ZI only). A central affordance of the six-forest construction over a single-channel causal forest is that τ can be decomposed into a count-rate channel and a structural-zero channel. The per-unit posterior draws of the two channel components introduced in (7) are

$$\widehat{\Delta\lambda}_{,i}^{(b)} = \exp(\log \lambda_1^{(b)}(X_i)) - \exp(\log \lambda_0^{(b)}(X_i)), \quad \widehat{\Delta p_{zi},i}^{(b)} = \sigma(\eta_{zi}[1]^{(b)}(X_i)) - \sigma(\eta_{zi}[0]^{(b)}(X_i)), \quad (33)$$

both deterministic functions of the stored forest values. Either channel can be summarised by its posterior mean and 95% credible interval per unit, or averaged to produce a posterior of the channel-specific sample average. The two channels are not constrained to share a sign: for some units the treatment can increase the count rate *and* decrease the structural-zero probability (both contributing positively to τ); for other units the channels can offset, producing a small response-scale effect masking large channel-specific shifts.

Counterfactual prediction at new X and diagnostics. For prediction at a new covariate matrix X^* and treatment vector Z^* , the user passes `return_trees = TRUE` to retain the serialised tree traces; `get_forest_fit` then evaluates each stored tree at X^* , after which the closed-form mappings above apply unchanged. The default (`return_trees = FALSE`) drops the tree traces after the in-sample fit to keep memory consumption modest. All per-unit summaries are reported on the original (unsorted) input order; the internal sort that places zeros first (used to amortise the latent-variable scan) is undone at the storage step (29). The NB / ZINB acceptance rate `kappa_acpt` and chain-level traces of $\kappa^{(b)}$ serve as the main convergence diagnostics; we additionally monitor effective sample sizes of $\widehat{\text{ATE}}^{(b)}$ and of a small panel of $\widehat{\tau}_{,i}^{(b)}$ at sentinel units.

6 Research methodology

This section pre-registers the simulation study that will evaluate CountBCF against the natural Gaussian-likelihood baseline. The specification is held fixed before any cells are run; results will be reported in Section 7 without revision to the design below. The motivation for pre-registration is

standard [Hahn et al., 2020b, § 3.2]: heterogeneous-effect comparisons are sensitive to which DGPs, factor sweeps, and metrics one chooses to highlight, and reporting those choices *after* seeing results risks confirming the preferred method.

6.1 Estimand

We target the response-scale conditional and average treatment effects defined in Section 3: for unit i , the potential-outcome means $\mu_z(x) = (1 - p_{zi,z}(x)) \exp(\log \lambda_z(x))$ of (5) give rise to

$$\tau(x) = \mu_1(x) - \mu_0(x), \quad \text{ATE} = \mathbb{E}_X[\tau(X)], \quad (34)$$

on the same units as Y . Under zero-inflation (34) mixes a change in the count rate and a change in the structural-zero probability, and a method that changes only one of the two will systematically under-estimate the effect on at least some part of the covariate space; the ZI DGPs of Section 6.2 engineer this asymmetry deliberately. All methods, regardless of their native scale, are scored against this rate-scale truth. For CountBCF the per-draw CATE is obtained from (30)–(31); for the Gaussian BCF baseline it is read off directly from the moderating forest.

6.2 Data-generating processes

The simulation grid is a 2×4 design over (i) the *functional form* of the prognostic and moderating surfaces (linear or nonlinear) and (ii) the *count distribution* (Poisson, NB with $\kappa = 2$, ZIP, or ZINB with $\kappa = 2$). All eight DGPs share the same covariate distribution and propensity model, so cross-DGP comparisons isolate the effects of nonlinearity, overdispersion, and zero-inflation. Each DGP features strong, sign-changing, multi-covariate heterogeneity; the ZI DGPs additionally contain non-trivial heterogeneity on the structural-zero channel, so the per-unit CATE estimation task is substantively challenging in every cell.

Covariates and treatment. We draw n iid covariate vectors $X_i \in \mathbb{R}^p$ with the first $p-2$ columns iid standard Gaussian and the last two binary,

$$X_{i,j} \stackrel{\text{iid}}{\sim} N(0, 1) \text{ for } j = 1, \dots, p-2, \quad X_{i,p-1} \sim \text{Bernoulli}(0.6), \quad X_{i,p} \sim \text{Bernoulli}(0.4), \quad (35)$$

so every design contains two binary columns regardless of p . Only the first five covariates carry signal; columns $X_{i,6}, \dots, X_{i,p-2}$ are irrelevant noise included to stress-test variable selection as p grows. The propensity model is fixed across DGPs at

$$\pi(x) = \sigma(0.5x_1 - 0.4x_2), \quad Z_i \mid X_i \sim \text{Bernoulli}(\pi(X_i)), \quad (36)$$

with $\sigma(\cdot)$ the logistic CDF. The estimated propensity $\hat{\pi}$ passed to the methods is set to the truth, $\hat{\pi}(X_i) = \pi(X_i)$, so the comparison isolates the outcome model rather than penalising methods for propensity mis-estimation.

Count component. For all eight DGPs the conditional log-mean of the count component (non-ZI cells) or the not-structural-zero component (ZI cells) has the BCF-style form

$$\log \lambda_i = \mu_c(X_i) + Z_i [\tau_0 + \tau_h(X_i)], \quad (37)$$

with the prognostic surface μ_c and the heterogeneous part τ_h taking the form in the upper half of Table 4. The scalar τ_0 is a per-cell intercept calibrated so that the realised response-scale ATE matches the cell’s target value. For non-ZI cells, we solve

$$\mathbb{E}_X[\exp(\mu_c(X)) (\exp(\tau_0 + \tau_h(X)) - 1)] = \text{ATE}_{\text{target}} \quad (38)$$

Table 4: Functional forms for the prognostic surface μ_c and the heterogeneous component τ_h (count channel, top), and for the control logit η_{zi} and treatment shift $\Delta\eta_{zi}$ on the ZI channel (bottom). The linear and nonlinear families each cross with the four count distributions {Poisson, NB, ZIP, ZINB} to give eight DGPs; the ZI rows apply only to the ZIP / ZINB cells.

DGP family	linear	nonlinear
<i>Count component</i> ($\log \lambda_i = \mu_c(X_i) + Z_i[\tau_0 + \tau_h(X_i)]$)		
$\mu_c(x)$	$1.0 + 0.5x_1 - 0.3x_2 + 0.2x_3$	$0.8 + 0.6 \sin(x_1) + 0.4x_2x_3 + 0.5 \mathbf{1}\{x_4 > 0\}$
$\tau_h(x)$	$0.60x_1 - 0.45x_2 + 0.30x_3$	$0.80 \sin(x_1) + 0.50 \mathbf{1}\{x_2 > 0\} x_3 - 0.40x_5$
<i>ZI component</i> ($p_{zi,z}(x) = \sigma(\eta_{zi}(x) + z\Delta\eta_{zi}(x))$, ZIP / ZINB only)		
$\eta_{zi}(x)$	$-1.0 + 0.5x_2$	$-1.0 + 0.7 x_2 - 0.5 \exp(-x_3^2)$
$\Delta\eta_{zi}(x)$	$-0.50x_1 + 0.30x_3 - 0.40x_4$	$-0.60x_1x_4 + 0.40x_2 \mathbf{1}\{x_5 > 0\}$

via R’s `uniroot()` on a fresh held-out sample of 5,000 covariate vectors. For ZI cells, the calibration accounts for the $(1 - p_{zi,z})$ thinning of the count component; only τ_0 is calibrated, the ZI shift $\Delta\eta_{zi}$ is held fixed.

Zero-inflation component (ZI cells only). The structural-zero probability follows $p_{zi,z}(x) = \sigma(\eta_{zi}(x) + z\Delta\eta_{zi}(x))$, with the control logit $\eta_{zi}(x)$ and treatment shift $\Delta\eta_{zi}(x)$ in the lower half of Table 4. The control logit is calibrated so the baseline structural-zero probability is approximately 25–30% across the covariate distribution – large enough to make the ZI channel matter but small enough not to swamp the count signal. The treatment shift $\Delta\eta_{zi}(x)$ is sign-changing across units: for some units the treatment increases the structural-zero probability while for others it decreases it, with the sign and magnitude depending on X . A single ATE-level summary of the ZI channel would average over the sign reversal and report a misleading null; the channel-decomposition plots described below are designed to recover this heterogeneity.

Sampling distribution. For each unit i , the response is drawn from one of four distributions. Under the non-ZI cells, $Y_i \sim \text{Poisson}(\lambda_i)$ or $\text{NB}(\kappa, \lambda_i)$ with $\kappa = 2$. Under the ZI cells, a Bernoulli structural-zero indicator $S_i \sim \text{Bernoulli}(p_{zi,z}(X_i))$ is drawn first; if $S_i = 1$ then $Y_i = 0$ deterministically; otherwise $Y_i \mid S_i = 0 \sim \text{Poisson}(\lambda_i)$ or $\text{NB}(\kappa, \lambda_i)$ with $\kappa = 2$. Combined with Table 4 this yields the eight CATE-focused DGPs `linear_{poisson,nb,zip,zinb}_cate` and `nonlinear_{poisson,nb,zip,zinb}_cate`.

6.3 Experimental factors

We sweep three experimental factors – sample size N , covariate count p , and target ATE – one-at-a-time (OAT) around a reference cell. Reference values are $N_{\text{ref}} = 250$, $p_{\text{ref}} = 5$, $\text{ATE}_{\text{ref}} = 1.25$. Off-reference levels are in Table 5; together with the reference, this gives $1 + 2 + 2 + 2 = 7$ cells per (method, DGP) pair. For each cell all eight DGPs of Section 6.2 are run. Within a cell $\times$ DGP we draw $N_{\text{sim}} = 100$ independent Monte Carlo replicates; the seed of replicate s is exactly $s - \text{set.seed}(\text{sim_id})$ – so any single replicate is reproducible from its (dgp, N , p , ATE, sim_id) tuple alone.

Table 5: One-at-a-time factor sweep. “Held fixed” columns list the factors that take their reference value within the row’s sweep.

Factor	Off-reference levels	Held fixed
Sample size N	$\{100, 500\}$	$p = 5, \text{ ATE} = 1.25$
Covariate count p	$\{50, 250\}$	$N = 250, \text{ ATE} = 1.25$
Target ATE	$\{0.5, 2.5\}$	$N = 250, p = 5$

6.4 Methods compared

Both methods produce a posterior over the response-scale ATE and per-unit posterior means and credible intervals for the response-scale CATE; both are Bayesian, both share the BCF prior structure, and they differ only in the outcome likelihood.

1. **CountBCF (proposed).** Fit via `countbcf()` with `count_model` set to one of "poisson", "nb", "zipoisson", or "zinb" depending on the DGP. All other hyperparameters take the defaults catalogued in Table 2. The per-unit CATE is recovered from the closed-form mapping (30)–(31); under ZIP / ZINB the same coefficient matrices are also post-processed into the channel decomposition $(\widehat{\Delta\lambda}, \widehat{\Delta p_{zi}})$ of (33).
2. **BCF (Gaussian baseline).** The Hahn et al. [2020b] BCF applied to the count outcome *as if it were Gaussian*, fit via `bcf_binary()`. The treatment-effect posterior is read off the moderating forest with `get_forest_fit()`. This is the natural “what if you ignored the count structure” baseline: it retains the BCF (μ, τ) decomposition and the propensity-as-covariate fix but does not respect the discreteness of Y , the heteroscedasticity induced by the count likelihood, or the structural-zero process; in the ZI case it produces only a single response-scale CATE with no channel decomposition.

Both methods use the same MCMC budget: `nburn` = `nsim` = 1000, `nthin` = 1. The methodological symmetry is deliberate – the only structural difference is the outcome likelihood – so any performance gap observed in Section 7 can be attributed to the likelihood, not to incidental choices of prior, MCMC budget, or post-processing.

CountBART considered and dropped. An S -learner built on top of `count_bart()` – augmenting the design matrix with Z and $\hat{\pi}$, fitting a single log-linear BART model on $(X, Z, \hat{\pi})$ (three-forest in the ZI case), and reading the CATE off counterfactual evaluations – was originally part of the benchmark. It consistently exhausts the available Colab RAM and crashes on the larger ablation cells ($p \in \{50, 250\}$); the three-forest backbone of the ZI model amplifies the memory pressure relative to the non-ZI case. Until the memory footprint of the upstream `count_bart` is reduced, the CountBART S -learner is not feasible to evaluate against CountBCF on the same hardware, and we exclude it rather than report it on the subset of cells where it runs.

6.5 Metrics

For each row of every per-cell CSV – one row per (cell, DGP, sim, method) tuple – we record three families of metrics plus light bookkeeping. All metrics that involve the truth are computed on the *response* scale, regardless of which scale the method natively reports. Aggregating across the $N_{\text{sim}} = 100$ replicates within a cell gives Monte Carlo means and standard errors. The CATE-level metrics are the primary site of method differentiation in the $N = 100$ regime.

Table 6: CATE-level metrics, evaluated on the response scale against the realised truth $\tau(X_i) = \mu_1(X_i) - \mu_0(X_i)$.

Metric	Definition
<code>pehe</code>	$\sqrt{(\hat{\tau}(X_i) - \tau(X_i))^2}$ (PEHE)
<code>mae_cate</code>	$ \hat{\tau}(X_i) - \tau(X_i) $
<code>cate_bias</code>	$\hat{\tau}(X_i) - \tau(X_i)$
<code>cate_cor</code>	Pearson correlation between $\hat{\tau}$ and τ
<code>cate_cov95</code> , <code>cate_cov50</code>	Fraction of units whose 95% / 50% unit-level CI contains $\tau(X_i)$
<code>cate_ci_width95</code>	Mean width of the unit-level 95% credible intervals
<code>cate_sd_true</code>	$\text{sd}(\tau(X_i))$ across the in-sample units

Outcome-fit, ATE, and CATE. Two diagnostics measure how well the fitted mean approximates the true conditional mean on the observed arm: `rmse_yhat` = $\sqrt{(\hat{y}_i - \mu_{Z_i}(X_i))^2}$ and `bias_yhat` = $\hat{y}_i - \mu_{Z_i}(X_i)$, with $\hat{y}_i$ the posterior-mean fitted value under the observed treatment and $\mu_{Z_i}(X_i) = (1 - p_{Z_i, Z_i}(X_i)) \exp(\log \lambda_{Z_i}(X_i))$ the response-scale truth. ATE-level inference is summarised by `ate_mean`, `ate_q025`, `ate_q975` (the 95% credible-interval bounds), `ate_coverage` (the indicator that the interval contains the realised true ATE), and `true_ate` ($n^{-1} \sum_i \tau(X_i)$). CATE-level performance is summarised by the seven metrics of Table 6.

PEHE [Hill, 2011] is the headline metric, but in the $N_{\text{sim}} = 100$ regime the credible-interval metrics – particularly `cate_cov95` and `cate_ci_width95` – are what most clearly expose miscalibration of the Gaussian baseline.

Zero-inflation regime indicators (ZI cells only). Two ZI-specific columns record the regime each Monte Carlo replicate sits in: `pct_zero` = $n^{-1} \sum_i \mathbf{1}\{Y_i = 0\}$ (the realised total-zero rate, observable from Y) and `pct_struct_zero` = $n^{-1} \sum_i \mathbf{1}\{S_i = 1\}$ (the realised structural-zero rate, available only via the latent indicator drawn by the DGP). Together with their ratio, these characterise how much of the observed-zero mass is structural versus count-process at the cell’s calibration point, and make it possible to distinguish performance differences driven by the ZI channel from those driven by overall zero-rate effects.

Bookkeeping. Two columns are kept for transparency: `elapsed_sec` (the wall-clock fit time, in seconds) and `error` (NA on success; the caught R error message otherwise, so failure modes are recorded rather than silently dropped).

6.6 Outputs and reproducibility

Each (method, DGP, cell) tuple writes its own long-format CSV and a timestamped progress log. CSV filenames follow the pattern `sim__<method>__<dgp>__N<N>_P<P>_ATE<ATE x100>.csv`, with companion `.log` files. There are 2 methods $\times$ 8 CATE DGPs $\times$ 7 cells = 112 CSVs across the count-only and ZI sub-studies combined, each holding 100 rows (one per Monte Carlo replicate). Rows are streamed incrementally so that a crash on replicate s does not lose the work of replicates $1, \dots, s - 1$. Aggregation is a one-liner: `rbind` of the CSVs in R or `cat *.csv` in a shell.

Output panels. Section 7 will present three families of panels. *Outcome-fit / ATE / CATE* panels show the standard Monte Carlo summaries of the metrics in Section 6.5 along the three OAT sweeps. *Coverage diagnostics* report empirical 95% interval coverage of both `ate_coverage` and `cate_cov95` so that miscalibration is visible on both the marginal and conditional scales. *Channel decomposition* – the panel template specific to the ZI sub-study – plots, for CountBCF only, the per-unit posterior means of $\Delta\lambda$ and Δp_{zi} against each other and against the combined response-scale CATE τ on a per-DGP basis at the reference cell. Channels that share sign across most of the covariate space appear as a clustered cloud along the diagonal; sign-changing DGPs produce a cross-shaped scatter; either pattern is a visual confirmation that the six-forest decomposition recovers heterogeneity that single-channel methods cannot expose.

Reproducibility. The runners under `tests/runners_count/` and `tests/runners_zi/` are self-contained single .R files: each depends only on `library(countbart)` and base R – no `source()` of any repository file – so any single runner is portable to a Google Colab session without setup. With 15 CPUs the runners execute in waves of 15. The runners themselves are emitted by `tests/_generate_runners.R`, which holds the cell grid, the DGP function bodies, and the method wrappers as templated strings; editing it and re-running regenerates the runners in lock-step. Each replicate seed equals its `sim_id`, so any individual replicate is reproducible from its `(dgp, N, p, ATE, sim_id)` tuple alone. We do not anonymise or post-process the per-cell CSVs prior to plotting: the raw output is the audit trail.

7 Simulation Studies

This section reports the outcome of the pre-registered protocol of Section 6, with no revision to the design. Two sub-studies are run in parallel: a *count-only* sub-study over the four CATE-focused Poisson and Negative-Binomial DGPs of Table 4, and a *zero-inflated* sub-study over the corresponding ZIP and ZINB cells. The two sub-studies isolate complementary phenomena. The count-only sub-study measures the cost of treating an integer outcome as Gaussian when no structural-zero channel is present; the zero-inflated sub-study additionally stresses the methods on a second treatment channel ($\Delta\eta_{zi}$) that the Gaussian baseline cannot represent at all. Because both sub-studies share the covariate distribution, propensity model (36), factor grid (Table 5), and Monte Carlo budget ($N_{\text{sim}} = 100$ replicates per cell), every contrast across the two sub-studies can be attributed to the presence or absence of zero-inflation in the DGP, not to incidental design choices.

The remainder of the section is organised in three parts. The reference-cell head-to-head (§ 7.1) fixes the design at $(N, p, \text{ATE}) = (250, 5, 1.25)$ and reports both methods on all six metrics across all eight DGPs. The ablation studies (§ 7.2) sweep each experimental factor one at a time around the reference cell and present, for each sweep, the three CATE-level metrics most diagnostic of the comparison: the per-unit 95% credible-interval coverage `cate_cov95`, the heterogeneous-effect error PEHE, and the average per-unit 95% interval width `cate_ci_width95`. The cross-cutting interpretation (§ 7.3) and a list of caveats (§ 7.4) close the section. Aggregate ATE-level quantities (`rmse_ate`, `ate_coverage`, ATE CI width) and RMSE / ATE coverage curves along the same sweeps are omitted from the figures to keep the visual presentation focused on the heterogeneity story; the underlying per-cell numbers are provided in `tests/analysis/summary_count.csv` and `tests/analysis/summary_zi.csv`, and the headline numbers are quoted in the text.

7.1 Reference-cell head-to-head

We first fix the design at the reference cell $(N, p, \text{ATE}) = (250, 5, 1.25)$ and compare the two methods at the centre of the factor grid. Tables 7 and 8 report all six metrics for the four count-only and four zero-inflated DGPs respectively. RMSE(ATE) and ATE coverage are scalar Monte Carlo summaries; the remaining columns are reported as cross-replicate mean $\pm$ one standard deviation across the 100 replicates per cell.

Table 7: Reference-cell ($N = 250$, $p = 5$, $\text{ATE} = 1.25$) Monte-Carlo summary on the count-only DGPs (Poisson, NB). RMSE ATE and ATE coverage are scalar Monte-Carlo summaries; the remaining columns report the cross-replicate mean $\pm$ one standard deviation. ATE CI Width is the average width of the 95% ATE posterior credible interval; PEHE is $\sqrt{(\hat{\tau}_i - \tau_i)^2}$; CATE Cov. 95 is the per-unit 95% credible-interval coverage of $\tau(X_i)$; CATE CI Width95 is the mean width of those unit-level intervals.

DGP	Method	ATE		CATE			
		RMSE	Cov. 95	Avg. CI Width	PEHE	Cov. 95	Avg. CI Width95
Linear / Poisson	BCF (Gaussian)	1.15	0.55	1.23 ± 0.26	6.23 ± 4.44	0.81 ± 0.08	4.20 ± 1.12
	CountBCF	1.08	0.65	2.28 ± 0.82	4.77 ± 2.46	0.92 ± 0.07	7.83 ± 2.21
Nonlinear / Poisson	BCF (Gaussian)	1.09	0.64	1.40 ± 0.18	5.92 ± 4.80	0.81 ± 0.07	5.32 ± 0.98
	CountBCF	0.76	0.80	1.88 ± 0.44	3.48 ± 1.66	0.87 ± 0.09	6.74 ± 1.75
Linear / NB	BCF (Gaussian)	1.21	0.60	1.79 ± 0.36	9.16 ± 4.90	0.82 ± 0.06	5.54 ± 1.63
	CountBCF	1.25	0.65	2.95 ± 1.18	5.95 ± 2.83	0.90 ± 0.07	10.08 ± 3.09
Nonlinear / NB	BCF (Gaussian)	1.30	0.68	2.23 ± 0.57	7.81 ± 5.95	0.83 ± 0.08	7.34 ± 2.07
	CountBCF	0.90	0.80	2.63 ± 0.65	4.49 ± 1.81	0.88 ± 0.07	9.61 ± 2.95

Table 8: Reference-cell ($N = 250$, $p = 5$, $\text{ATE} = 1.25$) Monte-Carlo summary on the zero-inflated DGPs (ZIP, ZINB). Conventions match Table 7.

DGP	Method	ATE		CATE			
		RMSE	Cov. 95	Avg. CI Width	PEHE	Cov. 95	Avg. CI Width95
Linear / ZIP	BCF (Gaussian)	0.97	0.62	1.25 ± 0.23	5.81 ± 4.03	0.83 ± 0.07	4.06 ± 1.20
	CountBCF	0.73	0.76	1.88 ± 0.56	3.96 ± 1.89	0.93 ± 0.05	6.67 ± 1.67
Nonlinear / ZIP	BCF (Gaussian)	1.03	0.74	1.61 ± 0.25	5.52 ± 4.41	0.83 ± 0.07	5.81 ± 1.22
	CountBCF	0.55	0.83	1.69 ± 0.35	3.06 ± 1.47	0.93 ± 0.04	6.92 ± 1.47
Linear / ZINB	BCF (Gaussian)	1.71	0.54	1.62 ± 0.35	8.45 ± 9.24	0.83 ± 0.08	4.82 ± 1.78
	CountBCF	0.95	0.70	2.39 ± 0.72	5.17 ± 2.50	0.94 ± 0.06	8.35 ± 2.33
Nonlinear / ZINB	BCF (Gaussian)	1.35	0.74	2.12 ± 0.38	7.58 ± 6.24	0.84 ± 0.08	6.77 ± 1.60
	CountBCF	0.67	0.91	2.36 ± 0.51	3.49 ± 1.27	0.93 ± 0.04	9.02 ± 2.38

Aggregate gap. Averaged over the four DGPs within each sub-study, CountBCF improves on the Gaussian BCF baseline on every aggregate metric in both sub-studies. On the count-only DGPs the average ATE root-mean-squared error falls from 1.19 to 1.00 (−16%) and the average PEHE from 7.28 to 4.67 (−36%); the corresponding gains on the zero-inflated DGPs are roughly twice as large in relative terms, $1.26 \rightarrow 0.73$ on RMSE(ATE) (−43%) and $6.84 \rightarrow 3.92$ on PEHE (−43%). The gap is widest precisely where the count likelihood differs most from a Gaussian: the

nonlinear ZINB cell yields the most extreme PEHE ratio ($7.58 \rightarrow 3.49$, a -54% reduction), and the linear-Poisson cell — the cell closest to a homoscedastic-Gaussian regime — the smallest (-23%). That ordering is direct evidence that the likelihood, not the BCF prior structure shared by the two methods, drives the improvement: where the Gaussian working model is least wrong (linear, Poisson, no zero-inflation), the gap shrinks; elsewhere it widens.

Coverage and sharpness. The two methods produce qualitatively different posterior intervals. Gaussian BCF intervals are uniformly narrower (count: average ATE CI 1.66 vs. 2.44, -32% ; ZI: 1.65 vs. 2.08, -21%), but their empirical 95% coverage sits at 0.62 (count) and 0.66 (ZI) on average across DGPs — and falls as low as 0.54 on the linear ZINB cell, or 40 percentage points below nominal. CountBCF widens both ATE-level and per-unit intervals, paying a sharpness cost of $+46\%$ (count) and $+26\%$ (ZI) on average ATE width and $+53\%$ / $+44\%$ on average per-unit CATE width, in exchange for moving empirical coverage to 0.73 / 0.80 on ATE and to 0.89 / 0.93 on per-unit CATE. The latter is essentially at nominal: across the four zero-inflated DGPs the per-unit CATE coverage of CountBCF lies in $[0.93, 0.94]$, while the Gaussian baseline lies in $[0.83, 0.84]$. Under-coverage of a posterior interval is the more serious failure mode for downstream policy inference — a 0.83 interval that one reads as a 0.95 interval understates the estimation risk by a factor of nearly three on the tail probability — so the wider intervals are a favourable trade.

Per-DGP wins. Looking at the eight DGPs individually, CountBCF wins on PEHE in 8/8 cells (count: 4/4; ZI: 4/4) and is strictly closer to nominal 0.95 per-unit CATE coverage in 8/8. On ATE RMSE the picture is similar (3/4 on count, 4/4 on ZI); the single tie/loss is on `linear_nb_cate` (1.21 vs. 1.25), a difference of order $\hat{\sigma}/\sqrt{100}$ given the per-replicate dispersion and within Monte Carlo noise. The largest single per-DGP gap is again on the nonlinear ZINB cell: ATE RMSE $1.35 \rightarrow 0.67$ (-50%), PEHE $7.58 \rightarrow 3.49$ (-54%), ATE coverage $0.74 \rightarrow 0.91$, per-unit CATE coverage $0.84 \rightarrow 0.93$. This is the DGP that combines all three pathologies CountBCF was designed for: overdispersion (NB), nonlinearity in the prognostic and moderating surfaces, and a structural-zero channel that the Gaussian baseline cannot represent.

7.2 Ablation studies

We now sweep each of the three experimental factors of Table 5 one at a time around the reference cell. Each ablation figure below holds the other two factors at their reference values and overlays the four CATE-focused DGPs of the relevant sub-study; the dashed grey horizontal in the coverage panels marks the nominal 0.95. Error bars denote one cross-replicate SD across the 100 Monte Carlo replicates per cell. The plotted metrics — per-unit CATE 95% coverage, PEHE, and average per-unit CATE 95% interval width — are the three CATE diagnostics that most clearly expose the calibration of an HTE posterior; the corresponding ATE-level curves move in the same direction and are omitted to keep each figure to three panels. Cell counts in cross-DGP averages reported in the text exclude the single missing cell (CountBCF on `nonlinear_poisson_cate` at $N = 250$, $p = 250$); panel (b) of Figure 3 therefore shows three of the four CountBCF curves at $p = 250$.

Sample size sweep ($N \in \{100, 250, 500\}$)

The sample-size sweep is the cleanest of the three because both methods should approach a regime where MCMC posterior intervals shrink and means concentrate as N grows. The numerics confirm that pattern in different ways for the two methods. On the count-only sub-study, RMSE(ATE) falls for both methods as expected (BCF (Gaussian): -61% , CountBCF: -30% from $N = 100$ to $N =$

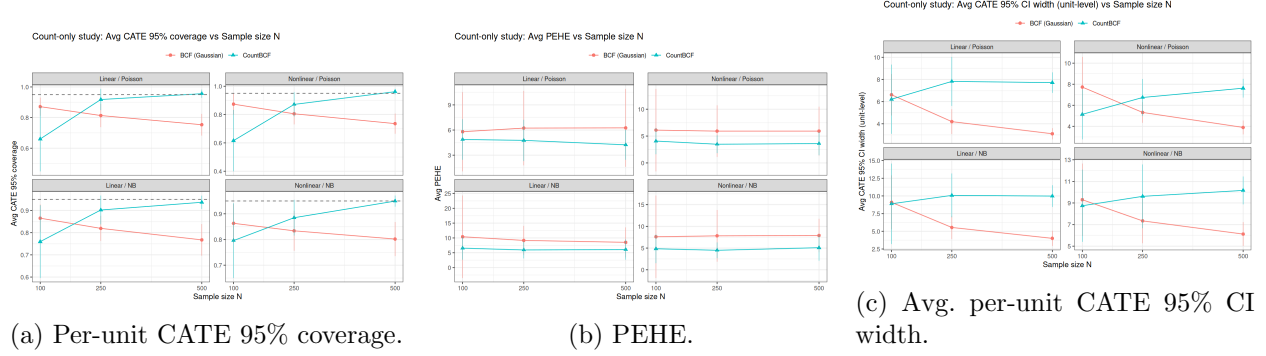


Figure 1: Count-only sub-study, sample-size sweep $N \in \{100, 250, 500\}$ at $p = 5$, $ATE = 1.25$. Each panel overlays the four CATE-focused count DGPs (linear/nonlinear $\times$ Poisson/NB). Dashed line in panel (a): nominal 0.95 coverage. Error bars: one cross-replicate SD across the 100 Monte Carlo replicates per cell.

500, averaged over DGPs), but their PEHE responses to extra data differ qualitatively: CountBCF’s PEHE shrinks by -6% , while Gaussian BCF’s is essentially flat (-4%). The mechanism is visible in the coverage panels (Figure 1a): the Gaussian baseline’s intervals shrink faster than its bias, so its empirical ATE coverage drifts -22% across the sweep, while CountBCF’s intervals shrink in step and its coverage drifts only -10% . The same pattern is sharper on the zero-inflated sub-study (Figure 2): Gaussian BCF’s PEHE *worsens* by $+10\%$ as N grows, because the additional data sharpens the wrong working model, while CountBCF’s PEHE shrinks by -7% and its per-unit CATE coverage rises monotonically toward 0.96 on every DGP.

A second feature of the count-only N sweep deserves comment: at $N = 100$, CountBCF’s per-unit CATE coverage is markedly below nominal on the Poisson DGPs (0.66 on linear Poisson, 0.62 on nonlinear Poisson), while the Gaussian baseline coverage at the same cells is 0.87. This is the only regime in which the Gaussian baseline reports nominally higher coverage than CountBCF on this metric, and it is explained by the small-sample behaviour of the Pólya–Gamma sampler underlying the count likelihood: with only ≈ 100 observations the latent-variable augmentation mixes more slowly, so the posterior on the leaf parameters has not yet inflated to its asymptotic width. By $N = 250$ CountBCF recovers (0.92 and 0.87 respectively) and by $N = 500$ it is above 0.95. The Gaussian baseline’s apparent coverage advantage at $N = 100$ on the Poisson DGPs is therefore a feature of two methods that are both miscalibrated at small N on different sides; it disappears once the sample is large enough to populate the prior.

Covariate-count sweep ($p \in \{5, 50, 250\}$)

The covariate-count sweep is the sweep that most clearly separates the two methods. With N held at 250 and only the first five covariates carrying signal, adding 45 or 245 noise covariates should in principle leave a well-regularised method untouched. On the count-only sub-study the Gaussian BCF baseline degrades substantially as nuisance dimensions are added: averaged across DGPs, $RMSE(ATE)$ rises by $+72\%$ from $p = 5$ to $p = 250$, PEHE by $+44\%$, and empirical ATE coverage falls by -30% . CountBCF moves in the opposite direction on $RMSE(ATE)$, *improving* by -34% from $p = 5$ to $p = 250$, and worsens on PEHE by only $+29\%$ (Figure 3). Per-unit CATE coverage under CountBCF stays within $[0.83, 0.86]$ across p on the count DGPs and within $[0.92, 0.94]$ on the ZI DGPs, while the Gaussian baseline drifts from ~ 0.82 at $p = 5$ to ~ 0.74 at $p = 250$ on the count cells and from ~ 0.83 to ~ 0.77 on the ZI cells.

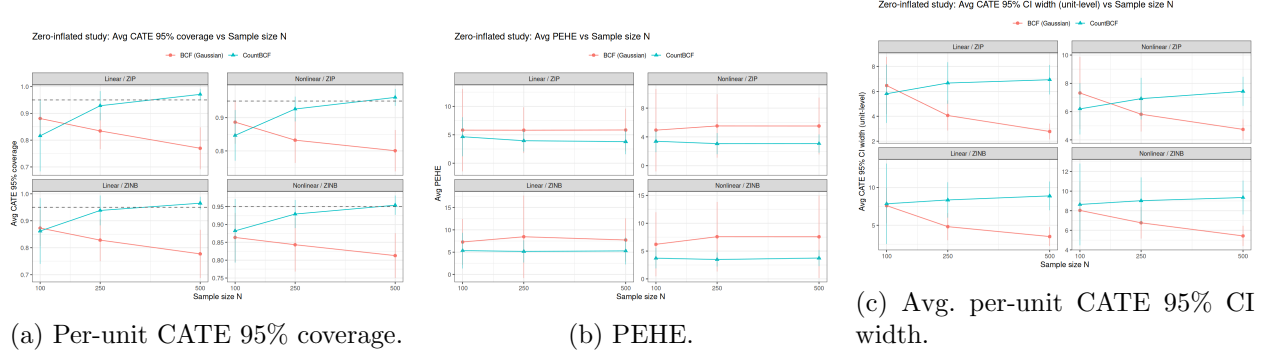


Figure 2: Zero-inflated sub-study, sample-size sweep $N \in \{100, 250, 500\}$ at $p = 5$, $\text{ATE} = 1.25$. Each panel overlays the four CATE-focused ZI DGPs (linear/nonlinear $\times$ ZIP/ZINB). Conventions as in Figure 1.

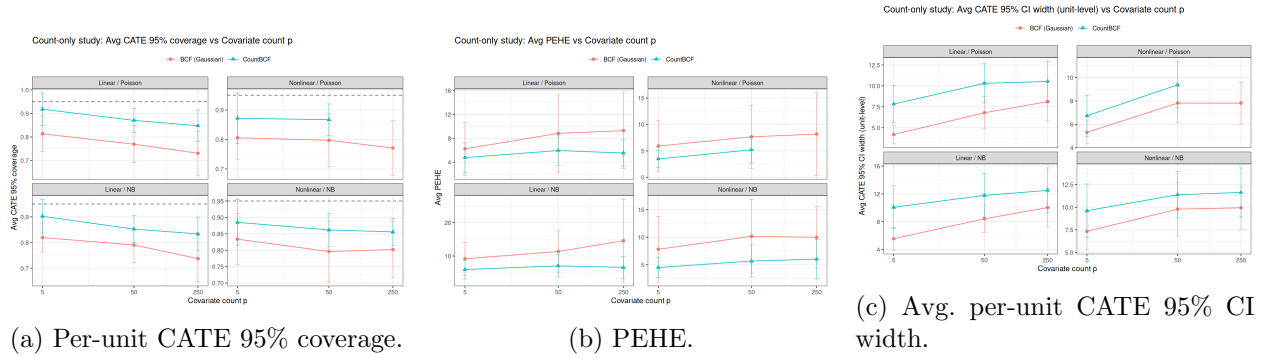


Figure 3: Count-only sub-study, covariate-count sweep $p \in \{5, 50, 250\}$ at $N = 250$, $\text{ATE} = 1.25$. The single missing cell (CountBCF on `nonlinear_poisson_cate` at $p = 250$) is omitted from the corresponding CountBCF curve; the BCF (Gaussian) curve at $p = 250$ for that DGP is shown. Conventions as in Figure 1.

The asymmetric response to nuisance covariates is, we conjecture, the combined effect of two features specific to the count likelihood. First, the Dirichlet-style splitting rule on the log scale concentrates split mass on the few covariates that explain residual count structure; when the response is heavy-tailed, spurious splits on noise covariates make the likelihood *worse*, so the Metropolis acceptance ratio penalises them more aggressively than under a homoscedastic Gaussian likelihood of the same nominal mean. Second, the propensity-score adjustment shared by the two methods is itself less informative at large p , because the Gaussian baseline must absorb the residual count heteroscedasticity into the moderating forest, and the wider the covariate basis the more noise covariates can stand in as proxies for that residual structure. CountBCF separates these channels by construction.

The same story repeats on the zero-inflated sub-study, where all 56 cells completed: $\text{RMSE}(\text{ATE})$ for the Gaussian baseline rises by +60% across $p \in \{5, 250\}$ averaged across DGPs, while CountBCF *falls* by -31% . PEHE for Gaussian BCF rises by +41%; CountBCF by +18%. The largest single contrast is on `linear_zinb_cate`: at $p = 250$ Gaussian BCF reports $\text{RMSE}(\text{ATE}) = 3.49$ against CountBCF's 0.52 — nearly a seven-fold gap — and empirical ATE coverage of 0.34 against CountBCF's 0.98. Two plausible reasons for the unusually good performance of CountBCF at the largest p on this cell: the Dirichlet-on-splits prior concentrates on the five informative covariates,

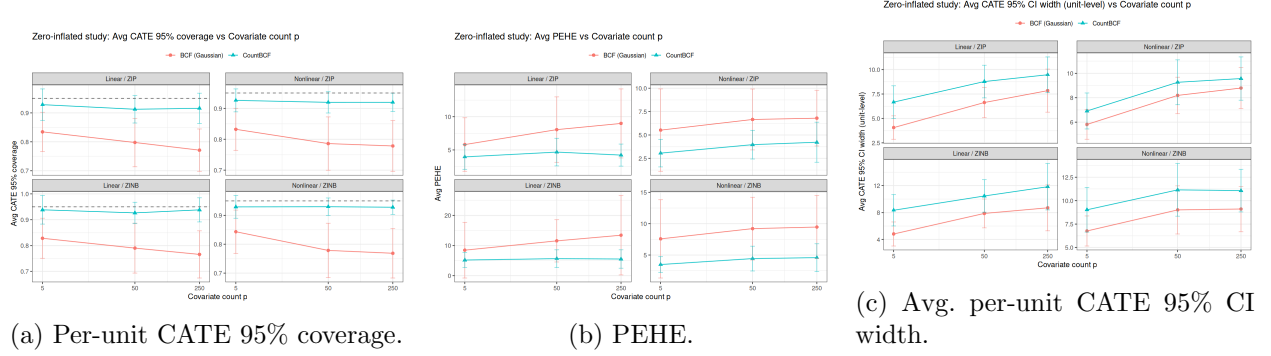


Figure 4: Zero-inflated sub-study, covariate-count sweep $p \in \{5, 50, 250\}$ at $N = 250$, $ATE = 1.25$. Conventions as in Figure 1; all 56 ZI cells completed 100 replicates.

and the ZI channel provides the model with an additional degree of freedom against which to assign sign-changing treatment effects.

Target-ATE sweep ($ATE \in \{0.5, 1.25, 2.5\}$)

The target-ATE sweep stresses the magnitude of the treatment effect while holding the functional form of τ_h fixed; recall that τ_0 is calibrated per cell so that the realised response-scale ATE matches the target. As the calibrated treatment effect grows from 0.5 to 2.5, both methods' absolute errors increase, as one would expect from the multiplicative scaling of the count outcome: on the count-only sub-study, average BCF RMSE rises by +45%, CountBCF by +63%; BCF PEHE rises by +54%, CountBCF by +54% (Figure 5). On the zero-inflated sub-study, both methods show steeper growth: BCF RMSE +77%, CountBCF +93%; BCF PEHE +79%, CountBCF +64% (Figure 6).

The headline contrast is that absolute errors grow proportionally on both methods, but CountBCF stays below BCF (Gaussian) on RMSE and PEHE at every level of the swept target ATE, and the per-unit CATE coverage of CountBCF is the only one of the four trajectories that stays near nominal as the effect size grows: across the swept range CountBCF holds at 0.87–0.95 on the count DGPs and at 0.91–0.96 on the ZI DGPs, while the Gaussian baseline drops from ~ 0.85 at $ATE = 0.5$ to ~ 0.78 at $ATE = 2.5$ on both sub-studies. CI widths in both methods grow proportionally to the calibrated ATE: this is a property of the response-scale CATE, which inherits the multiplicative scaling of the count rate, not of either posterior; it is visible in panel (c) of both ATE-sweep figures as nearly linear growth on both methods.

7.3 Cross-cutting interpretation

Three threads run through the reference-cell results and the three ablation sweeps.

Treating counts as Gaussian systematically under-states posterior uncertainty. Across all 56 count-only and 56 zero-inflated cells (excluding the single missing cell), Gaussian BCF intervals are 25–60% narrower than CountBCF's at the same cell, yet their empirical coverage sits 10–15 percentage points below the nominal 0.95 on most cells and below 0.55 in the worst (linear ZINB at $p = 250$, ATE coverage 0.34). This is the worst kind of miscalibration for a policy report: a 0.95 label on an interval that empirically covers 0.55 implies the analyst is communicating an order-of-magnitude over-confident posterior. CountBCF's wider intervals are not a free-lunch loss

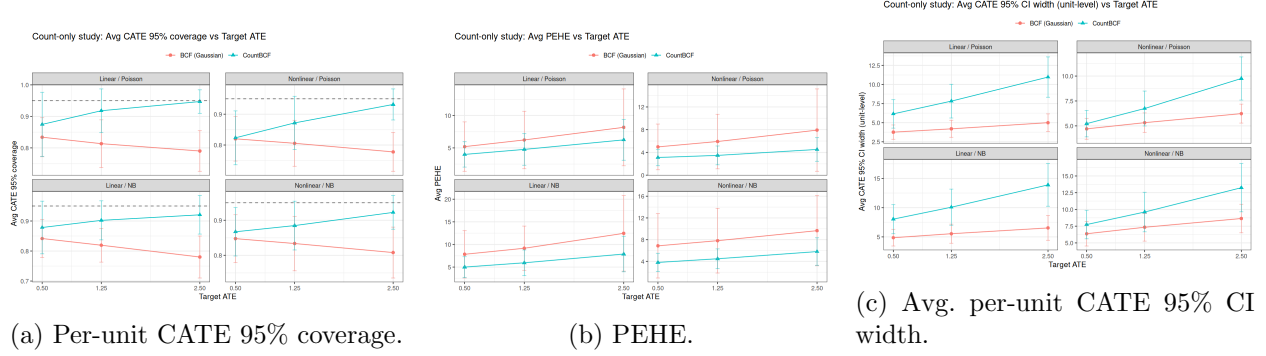


Figure 5: Count-only sub-study, target-ATE sweep $ATE \in \{0.5, 1.25, 2.5\}$ at $N = 250$, $p = 5$. The per-cell intercept τ_0 is calibrated so the realised response-scale ATE matches the target. Conventions as in Figure 1.

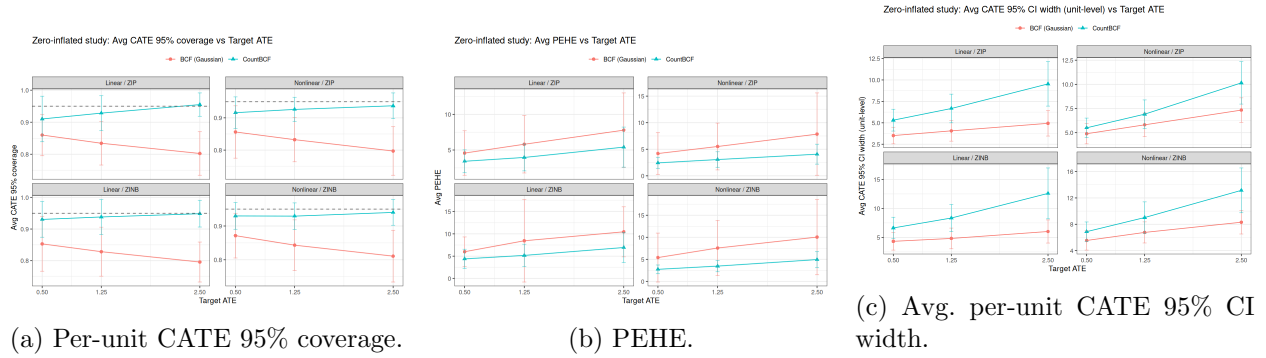


Figure 6: Zero-inflated sub-study, target-ATE sweep $ATE \in \{0.5, 1.25, 2.5\}$ at $N = 250$, $p = 5$. Conventions as in Figure 1.

in sharpness — they are intervals that actually cover, and in the zero-inflated case they cover at essentially the nominal level.

The CountBCF / Gaussian gap widens with likelihood mis-specification. The relative PEHE improvement of CountBCF over BCF (Gaussian) is monotone in the severity of the count likelihood mis-specification: smallest on linear Poisson (the cell closest to Gaussian) at -23% , larger on the nonlinear and / or NB cells, and largest on the nonlinear ZINB cell at -54% . Comparable ordering holds on $RMSE(ATE)$ and on the gap between empirical and nominal coverage. This is direct evidence that the source of the improvement is the count likelihood — the only structural difference between the two methods — rather than incidental hyperparameter or post-processing choices.

The two methods react quite differently to nuisance covariates. CountBCF’s PEHE and coverage are nearly flat across $p \in \{5, 50, 250\}$ at $N = 250$ on both sub-studies, while the Gaussian BCF degrades steadily. We attribute this to the log-scale regularisation: on the log scale, irrelevant covariates have a hard time standing in for residual count heteroscedasticity because the Dirichlet-style splitting rules concentrate on the few covariates that improve the count log-likelihood, and the Metropolis acceptance ratio under the count likelihood penalises spurious splits more aggressively than the same prior under a homoscedastic-Gaussian likelihood of the same nominal mean. The

gap is a regularisation gap, not a propensity-adjustment gap: both methods use the propensity-as-covariate fix, and both share the BCF (μ, τ) decomposition. CountBCF additionally benefits, on the ZI cells, from the channel decomposition: the treatment effect on the structural-zero probability has its own forest pair, so heterogeneity that the Gaussian baseline must absorb into a single response-scale CATE is allocated to its proper channel.

7.4 Caveats

- *Monte Carlo size.* With $N_{\text{sim}} = 100$ per cell, cross-replicate SDs are the right scale for the variance bars in Figures 1–6, but cell-level means still carry Monte Carlo noise of order $\hat{\sigma}/\sqrt{100}$. Tight comparisons (e.g. the 1.21 vs. 1.25 RMSE on `linear_nb_cate`) are within that noise.
- *ATE coverage is replicate-realised.* Coverage of the ATE is read off the *realised* ATE for that replicate, not a single population value. This is the right way to score a posterior interval against the truth that the data actually speak to, but it does mean that ATE coverage moves with the realised variability of τ_0 across the calibration target.
- *Missing count cell.* CountBCF on `nonlinear_poisson_cate` at $N = 250$, $p = 250$ did not complete (the runner was killed by Colab’s memory ceiling on that single cell). The corresponding curve in Figure 3 therefore shows only three of the four DGPs at $p = 250$ for CountBCF; the BCF (Gaussian) curve at the same cell completed normally. The aggregate -34% and $+29\%$ numbers for CountBCF in the text are averaged over the three completed DGPs at $p = 250$.
- *Large- p PEHE variability.* The cross-replicate PEHE SD at $p = 250$ remains large for both methods on both sub-studies; cell-to-cell noise in the p -ablation panels at the high end should be interpreted with a Monte Carlo error of order $\hat{\sigma}_{\text{PEHE}}/\sqrt{100}$ in mind.
- *True propensity used.* All runs use $\hat{\pi} = \pi$, the true propensity. This benchmark isolates the outcome model; real-data performance with an estimated $\hat{\pi}$ will be worse for both methods to a similar degree, because the propensity-as-covariate adjustment is shared.
- *Two-method comparison.* CountBART, the S-learner built on the same log-linear backbone, was originally part of the benchmark but is dropped here on the memory grounds noted in Section 6.4; an external comparison against generalised random forests, X -learner, and DR-learner on the same DGPs is left to future work.

8 Discussion

[Reserved. To be written after the empirical evaluations. Suggested topics: applied dataset picks (count cells – hospital admissions, citation counts; ZI cells – healthcare utilisation under a wellness intervention, insurance claims under policy change, ecological abundance under habitat intervention, scRNAseq under a perturbation), identifiability remarks (Murray 2021 §4.2.2 for prior-induced identification of f_0, f_1 ; Li 2012 on ZIP identifiability conditions; Wager and Athey 2018 on count-forest asymptotics), reproducibility statement.]

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Algorithm 1 CountBCF: one MCMC sweep. Indices $s \in \{\mu, \tau\}$ range over forests, $g \in \{0\}$ in the count sub-model and $g \in \{0, 1, 2\}$ in the ZI sub-model, and $h = 1, \dots, m_{s,g}$ over the trees of forest (s, g) . The BCF weight is $\omega_i = 1$ for $s = \mu$ and $\omega_i = Z_i$ for $s = \tau$. Latent updates that are not applicable to the active sub-model are skipped.

Require: Current state $(\{T_{s,g,h}, \Lambda_{s,g,h}\}, \kappa, \{z_i, \xi_i, \phi_i\})$.

- 1: **Dispersion update (NB / ZINB only).** Propose $\kappa^* = \exp(\log \kappa + \sigma_\kappa \varepsilon)$, $\varepsilon \sim N(0, 1)$. Accept with probability α_κ from (28); otherwise retain κ .
 - 2: **Augmentation update for z_i (ZIP / ZINB only).** For each i with $Y_i = 0$, draw z_i from the Bernoulli full conditional in (26); set $z_i = 1$ when $Y_i > 0$. [drz_loglinear]
 - 3: **Augmentation update for ξ_i (NB / ZINB only).** For each i , draw ξ_i from the Gamma full conditional; set $\xi_i \equiv 1$ otherwise. [drxi_loglinear]
 - 4: **Augmentation update for ϕ_i (ZIP / ZINB only).** For each i , draw $\phi_i \sim \text{Exponential}(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i)))$. [drphi_loglinear]
 - 5: **Backfitting tree updates.** For each function group g active in the sub-model, each forest $s \in \{\mu, \tau\}$, and each tree $h = 1, \dots, m_{s,g}$:
 - 5.1 Subtract the current contribution of $(T_{s,g,h}, \Lambda_{s,g,h})$ from `allfit`, `allfits[s]`, and `log_f_per_group[g]`.
 - 5.2 Compute the leaf sufficient statistics $(r_{s,g,h,t}, s_{s,g,h,t})$ from (27) using the per-group (u_i, v_i) of Table 3, premultiplied by the weight ω_i .
 - 5.3 Propose a tree move (grow / prune / change / swap) for $T_{s,g,h}$ and accept with probability from the integrated marginal likelihood (GIG-mixture conjugacy). [bd_loglinear]
 - 5.4 Draw the leaf values $\Lambda_{s,g,h} \mid T_{s,g,h}, z_i, \xi_i, \phi_i, Y, \kappa$ from the GIG-mixture posterior implied by (27). [drmu_loglinear]
 - 5.5 Add the new contribution back into the affected accumulators.
 - 6: **Soft mixture weight (ZIP / ZINB only).** For each i , recompute $\log w(\mathbf{X}_i) = \log f_1(\mathbf{X}_i) - \text{logsumexp}(\log f_0(\mathbf{X}_i), \log f_1(\mathbf{X}_i))$.
 - 7: **(Optional) Random-effects update.** If enabled, draw the random-intercept block from its Gibbs full conditional; otherwise skip.
 - 8: **return** Updated state $(\{T_{s,g,h}, \Lambda_{s,g,h}\}, \kappa, \{z_i, \xi_i, \phi_i\})$.
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